

# On the spum and sum-diameter of paths

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## ABSTRACT

In a sum graph, the vertices are labeled with distinct positive integers, and two vertices are adjacent if the sum of their labels is equal to the label of another vertex. In 1990, Harary showed that not all graphs  $G$  can be labeled as a sum graph but the union of  $G$  and at least some  $\sigma(G)$  isolated vertices can be. The spum of a graph  $G$  is defined as the minimum difference between the largest and smallest labels of a sum graph that consists of the union of  $G$  and exactly  $\sigma(G)$  isolated vertices. More recently, Li introduced the sum-diameter of a graph  $G$ , which modifies the definition of spum by removing the requirement that the number of isolated vertices must be  $\sigma(G)$ . In this paper, we settle conjectures by Singla, Tiwari, and Tripathi and a conjecture by Li by evaluating the spum and the sum-diameter of paths.

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## 1. Introduction

In 1990, Harary [2] defined the *sum graph*  $G(V, E)$  of  $L \subseteq \mathbb{N}$  to be given by  $V = L$  and  $(u, v) \in E$  if  $u + v \in L$  (see for example, Fig. 1). Not every graph  $G$  is a sum graph of some set  $L$ ; for example, no connected  $G$  is a sum graph, because in a sum graph, the vertex with the largest label must be isolated. Yet, Harary showed [2] that any graph  $G$  in union with at least a *sum number*  $\sigma(G)$  of isolated vertices is a sum graph of some set  $L$ . Henceforth, let  $I_n$  be a set of  $n$  isolated vertices for  $n \geq 1$ .

Shortly after, Goodell, Beveridge, Gallagher, Goodwin, Gyori, and Joseph [1] defined the *spum*  $\text{spum}(G)$  of a graph  $G$  as the minimum difference between the largest and the smallest labels in  $L$ , for which  $G \cup I_{\sigma(G)}$  is a sum graph of  $L$ . In the same

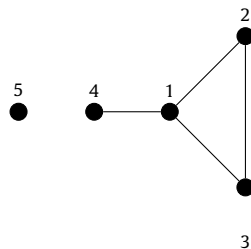


Fig. 1. The sum graph of  $\{1, 2, 3, 4, 5\}$ .

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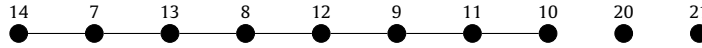


Fig. 2. A sum graph that demonstrates  $\text{sd}(P_8) \leq 14$ .

paper, Goodell et al. [1] evaluated  $\text{spum}(K_n)$ . More recently, in 2021, Singla, Tiwari, and Tripathi [5] evaluated  $\text{spum}(K_{1,n})$  and  $\text{spum}(K_{n,n})$ . Our first main result is Theorem 3.26, where we settle a conjecture that was originally proposed by Singla, Tiwari, and Tripathi [5, Conjecture 7.1] and subsequently modified by Li [4, Conjecture 3.4] on  $\text{spum}(P_n)$ , where  $P_n$  is a path of  $n$  vertices.

In 1994, Harary [3] extended the notion of a sum graph; he defined the *integral sum graph*  $G(V, E)$  of  $L \subseteq \mathbb{Z}$  to be given by  $V = L$  and  $(u, v) \in E$  if  $u + v \in L$ . He then defined the *integral sum number*  $\zeta(G)$  as the minimum number of vertices for which  $G \cup I_{\zeta(G)}$  is an integral sum graph. In 2021, Singla et al. [5] extended the notion of  $\text{spum}$ ; they defined the *integral spum*  $\text{ispum}(G)$  of a graph  $G$  as the minimum difference between the largest and the smallest labels in  $L$ , for which  $G \cup I_{\zeta(G)}$  is an integral sum graph of  $L$ . In the same paper, Singla et al. [5] evaluated  $\text{ispum}(K_n)$ ,  $\text{ispum}(K_{1,n})$ , and  $\text{ispum}(K_{n,n})$ . Our next main result Theorem 4.21 is an improvement on the best known lower bound on  $\text{ispum}(P_n)$  in [5].

Last year, Li [4] introduced the more natural *sum diameter*  $\text{sd}(G)$  of a graph  $G$  as the minimum difference between the largest and the smallest labels of  $L$ , for which  $G \cup I_m$  is a sum graph of  $L$  for any  $m \geq \sigma(G)$ .

**Remark.** Although a priori, it is not clear that adding more vertices reduces the range of its labels, there exist graphs  $G$  for which  $\text{sd}(G) < \text{spum}(G)$ . For example, while  $\text{spum}(P_8) = 15$  by Theorem 3.26, Fig. 2 shows that  $\text{sd}(P_8) \leq 14$ .

Our next main result is Theorem 5.6. We settle a conjecture by Li [4, Conjecture 9.5] on  $\text{sd}(P_n)$ .

Li [4] also introduced the *integral sum diameter*  $\text{isd}(G)$  of a graph  $G$  as the minimum difference between the largest and the smallest labels of  $L$ , for which  $G \cup I_m$  is an integral sum graph of  $L$  for any  $m \geq \zeta(G)$ . Our last main result is Theorem 4.32. We evaluate  $\text{isd}(P_n)$  for  $n \geq 3$ .

The rest of the paper is as follows. In Section 2, we establish preliminaries. In Section 3, we prove Theorem 3.26. In Section 4, we prove Theorems 4.21 and 4.32. In Section 5, we prove Theorem 5.6.

## 2. Preliminaries

For  $L \subseteq \mathbb{Z}$ , let  $L - a = \{\ell - a \mid \ell \in L\}$ , and  $L + a = \{\ell + a \mid \ell \in L\}$ . Next, let  $\text{range}(L) = \max(L) - \min(L)$ . Furthermore, let  $N_L(\ell) = \{a \in L \mid a \neq \ell \text{ and } \ell + a \in L\}$ . We remove the subscript  $L$  and write  $N_L(\ell)$  as  $N(\ell)$  when understood from context. Next, let  $L_i \subseteq L$  be the  $i$  smallest numbers in  $L$ . For  $L = \{a_1, a_2, \dots, a_n\} \subseteq \mathbb{Z}$  such that  $a_1 < a_2 < \dots < a_n$ , say that  $a_i$  and  $a_{i+1}$  are *consecutive* in  $L$ .

Let the maximal and minimal degree of a graph  $G$  be  $\Delta_G$  and  $\delta_G$ , respectively. We omit the subscript when it is clear which  $G$  is being referred to.

In [2] and [3] respectively, Harary evaluated  $\sigma(P_n)$  and  $\zeta(P_n)$ .

**Lemma 2.1** ([2]). For  $n \geq 1$ , it holds that  $\sigma(P_n) = 1$ .

**Lemma 2.2** ([3, Theorem 3.1]). For  $n \geq 1$ , it holds that  $\zeta(P_n) = 0$ .

## 3. The spum of paths

For this section, let  $G \cup I_{\sigma(G)}$  be a sum graph of  $L$  that satisfies  $\text{range}(L) = \text{spum}(G)$ . Let  $S = \{a_1, a_2, \dots, a_n\}$  be labels of the vertices  $V$  of  $G$  such that  $a_1 < a_2 < \dots < a_n$ . Next, let

$$T = [a_1, a_n] \setminus S,$$

$$M = (S \setminus [a_1, 2a_1]) - a_1, \text{ and}$$

$$J = (S \setminus [a_1, 3a_1]) - 2a_1.$$

We say that  $L$  is *tight* if  $T \subseteq M$  and  $k$ -*tight* if  $|T \setminus M| = k$ . Let  $m = a_1 - |S \cap [a_1, 2a_1]|$ . Then, because  $|S \cap [a_1, 2a_1]| = a_1 - m$  and  $S \sqcup T = [a_1, a_n]$ , it follows that

$$|T \cap [a_1, 2a_1]| = |[a_1, 2a_1]| - (a_1 - m) = m + 1. \quad (3.1)$$

Finally, let  $\epsilon = 1$  if  $(a_1, a_n) \in E$  is an edge of  $G$ , and  $\epsilon = 0$  otherwise.

### 3.1. The inequality $\text{spum}(P_n) \geq 2n - 2$ for $n \geq 7$

We first cite the best known bounds on  $\text{spum}(P_n)$ .

**Theorem 3.1** ([4, Theorem 3.1]). For  $n \geq 7$ , it holds that

$$2n - 2 \leq \text{spum}(P_n) \leq \begin{cases} 2n + 1 & \text{if } n \text{ is odd} \\ 2n - 1 & \text{if } n \text{ is even.} \end{cases}$$

We first generalize and correct an error in the original presentation of Claim 2 in [5].

**Lemma 3.2.** If  $G \cup I_i$  is a sum graph of  $L$  and  $2a_1 \in L$ , then

$$\text{range}(L) \geq 3n - a_1 - 3 + m - 4\Delta + \delta + 2\epsilon.$$

**Proof.** First, because  $|N(a_1)| \leq \Delta$ , at most  $\Delta - \epsilon$  pairs of labels in  $S \setminus \{a_1\}$  differ by  $a_1$ . As  $[a_1, 2a_1] \subseteq S \sqcup T = [a_1, a_n]$ , by Equation (3.1), it holds that

$$|S \cap [2a_1 + 1, 3a_1]| \leq |T \cap [a_1 + 1, 2a_1]| + \Delta - \epsilon = \Delta + m + 1 - \epsilon. \quad (3.2)$$

Because  $a_n \notin M$ , we have that  $S \cap M \subseteq N(a_1) \setminus a_n$ , so

$$|S \cap M| \leq |N(a_{r+1})| - \epsilon \leq \Delta - \epsilon. \quad (3.3)$$

Likewise, because  $M \cap J \subseteq N(a_1) \setminus a_n$ , it follows that

$$|M \cap J| \leq |N(a_{r+1})| - \epsilon \leq \Delta - \epsilon. \quad (3.4)$$

Similarly, because  $S \cap J \subseteq N(2a_1) \cup 2a_1$ , it follows that

$$|S \cap J| \leq |N(2a_1)| + 1 \leq \Delta + 1. \quad (3.5)$$

Notice that because  $|M| = |S \setminus [a_1, 2a_1]|$ , it follows that

$$|M| = |S| - |S \cap [a_1, 2a_1]| = n - (a_1 - m). \quad (3.6)$$

Because  $M \subseteq S \sqcup T = [a_1, a_n]$ , from Equations (3.3) and (3.6),

$$|T \cap M| = |M| - |S \cap M| \geq n - (a_1 - m) - (\Delta - \epsilon). \quad (3.7)$$

Now, from Equation (3.2),

$$\begin{aligned} |J| &= |S| - |S \cap [a_1, 2a_1]| - |S \cap [2a_1 + 1, 3a_1]| \\ &\geq n - (a_1 - m) - (\Delta + m + 1 - \epsilon) \\ &= n - a_1 - 1 - \Delta + \epsilon. \end{aligned} \quad (3.8)$$

As  $J \subseteq S \sqcup T = [a_1, a_n]$ , Equations (3.5) and (3.8) imply

$$\begin{aligned} |T \cap J| &= |J| - |S \cap J| \\ &\geq n - a_1 - 1 - \Delta + \epsilon - (\Delta + 1) \\ &\geq n - a_1 - 2 - 2\Delta + \epsilon. \end{aligned} \quad (3.9)$$

Lastly, from Equations (3.4), (3.7), and (3.9),

$$\begin{aligned} |T| &\geq |T \cap M| + |T \cap J| - |T \cap M \cap J| \geq |T \cap M| + |T \cap J| - |M \cap J| \\ &= (n - a_1 + m - \Delta + \epsilon) + (n - a_1 - 2 - 2\Delta + \epsilon) - (\Delta - \epsilon) \\ &= 2n - 4\Delta + 3\epsilon - 2a_1 + m - 2. \end{aligned} \quad (3.10)$$

Because  $|T| = (a_n - a_1 + 1) - n$ , from Equation (3.10),

$$a_n \geq 3n - a_1 - 3 + m - 4\Delta + 3\epsilon. \quad (3.11)$$

Now, because  $\max(N(a_n)) \geq a_1 + \delta - \epsilon$ , from Equation (3.11),

$$\text{range}(L) \geq (a_n + a_1 + \delta - \epsilon) - a_1 \geq 3n - a_1 - 3 + m - 4\Delta + \delta + 2\epsilon. \quad \square \quad (3.12)$$

**Remark.** The proof of Claim 2 in [5] claimed that  $|S \cap J| \leq \Delta$ , but it is possible that  $|S \cap J| = \Delta + 1$  if  $2a_1 \in S \cap J$ .

By specifying  $G = P_n$  in Lemma 3.2, we arrive at the following corollary.

**Corollary 3.3.** *If  $2a_1 \in S$ , then  $\text{spum}(P_n) \geq 3n - a_1 - 11 + 2\epsilon$ .*

3.2. *The inequality  $\text{spum}(P_n) \geq 2n - 1$  for  $n \geq 8$*

For the remainder of Section 3, let  $G = P_n \cup I_{\sigma(P_n)} = P_n \cup I_1$  from Lemma 2.1. In addition, let  $\{a_{n+1}\} = L \setminus S$ . Let  $b_i = a_{i+1} - a_i$  for  $1 \leq i \leq n$ , and  $c_\ell$  be the number of  $1 \leq i \leq n$  such that  $b_i = \ell$ . We first generalize [4, Lemma 2.5].

**Lemma 3.4.** *If  $L$  is  $k$ -tight, then  $2n - 3 + m + k + \epsilon \leq a_n \leq \text{range}(L)$ .*

**Proof.** By letting  $G = P_n$  in Equation (3.7),

$$|T \cap M| \geq n - a_1 + m - 2 + \epsilon. \quad (3.13)$$

Now, because  $|T \setminus M| = k$  and  $|T| = (a_n - a_1 + 1) - n$ , from Equation (3.13),

$$(a_n - a_1 + 1) - n = |T| = |T \cap M| + k \geq n - a_1 + m - 2 + \epsilon + k. \quad (3.14)$$

The statement of the lower bound of  $a_n$  now follows from rearranging Equation (3.14). In addition, because  $a_{n+1} \geq a_n + a_1$ , it follows that  $a_n \leq a_{n+1} - a_1 = \text{range}(L)$ .  $\square$

An immediate corollary of Lemma 3.4 is a lower bound for  $\text{range}(L)$ .

**Corollary 3.5.** *If  $L$  is  $k$ -tight, then  $\text{range}(L) \geq 2n - 2 + m + k$ .*

**Proof.** Because  $\max(N(a_n)) \geq a_1 + 1 - \epsilon$ , from Lemma 3.4,

$$\text{range}(L) \geq a_n + a_1 + 1 - \epsilon - a_1 \geq 2n - 2 + m + k. \quad \square \quad (3.15)$$

Next, we give an upper bound on  $a_1$  for  $k$ -tight  $L$ .

**Lemma 3.6.** *If  $L$  is  $k$ -tight, then  $a_1 \leq k + 2$ .*

**Proof.** Suppose otherwise. If  $M \neq \emptyset$ , then

$$\max(M) = a_n - a_1 \leq a_n - k - 3, \quad (3.16)$$

so regardless of whether or not  $M = \emptyset$ , it must hold that

$$[a_n - k - 2, a_n] \cap M = \emptyset. \quad (3.17)$$

Now, from Equation (3.17),

$$[a_n - k - 2, a_n] \cap T \subseteq T \setminus M. \quad (3.18)$$

Because  $S \sqcup T = [a_1, a_n]$  and  $L$  is  $k$ -tight, it holds from Equation (3.18) that

$$\begin{aligned} |S \cap [a_n - k - 2, a_n]| &= |[a_n - k - 2, a_n]| - |[a_n - k - 2, a_n] \cap T| \\ &\geq |[a_n - k - 2, a_n]| - |T \setminus M| \\ &= k + 3 - k = 3. \end{aligned} \quad (3.19)$$

Now, because  $\sigma(G) = 1$  and  $a_1 \geq k + 3$ , all vertices with labels in  $S \cap [a_n - k - 2, a_n]$  have one neighbor. However, exactly two vertices in  $P_n$  have one neighbor. Thus,  $a_1 \leq k + 2$ .  $\square$

Next, we derive an upper bound for  $a_n$ . To do so, we define

$$X = (S \setminus a_1) - a_1,$$

$$Y = (S \setminus [a_1, a_2]) - a_2, \text{ and}$$

$$Z = [1, a_n] \setminus S.$$

In addition, let  $\mu = |N(a_1) \setminus X| + |N(a_2) \setminus Y|$ .

**Lemma 3.7.** If  $2a_1 \notin L$ , then  $|Z \setminus X| + |Z \setminus Y| \leq 2a_n - 4n + 8 - \mu$ .

**Proof.** First, because  $|N(a_1)| \leq 2$  and  $2a_1 \notin L$ ,

$$|X \cap S| \leq 2 - |N(a_1) \setminus X|. \quad (3.20)$$

Furthermore, because  $|N(a_2)| \leq 2$  and it is possible that  $a_2 \in Y \cap S$ ,

$$|Y \cap S| \leq 3 - |N(a_2) \setminus Y|. \quad (3.21)$$

Because  $X, Y \subseteq Z \cup S = [1, a_n]$ , from Equations (3.20) and (3.21),

$$|X \cap Z| = |X| - |X \cap S| \geq (n-1) - (2 - |N(a_1) \setminus X|) = n-3 + |N(a_1) \setminus X|, \quad (3.22)$$

and

$$|Y \cap Z| = |Y| - |Y \cap S| \geq (n-2) - (3 - |N(a_2) \setminus Y|) = n-5 + |N(a_2) \setminus Y|. \quad (3.23)$$

Next, because  $|Z| = a_n - n$ , from Equations (3.22) and (3.23), it holds that

$$|Z \setminus X| = |Z| - |Z \cap X| \leq (a_n - n) - n + 3 - |N(a_1) \setminus X|, \quad (3.24)$$

and

$$|Z \setminus Y| = |Z| - |Z \cap Y| \leq (a_n - n) - n + 5 - |N(a_2) \setminus Y|. \quad (3.25)$$

Finally, the statement of the lemma follows from adding Equations (3.24) and (3.25).  $\square$

Next, we use Lemma 3.7 to bound  $n$  from above when  $a_1 + 1 = a_2$ . In the following proof, for some  $1 \leq i \leq j \leq n$ , we say that  $[a_i, a_j]$  is a *run* if  $[a_i, a_j] \subseteq S$  and  $\{a_i - 1, a_j + 1\} \cap S = \emptyset$ . Note that  $S = \bigsqcup_{i=1}^t R_i$ , where  $R_i$  are runs.

**Lemma 3.8.** If  $2a_1 \notin S$  and  $a_1 + 1 = a_2$ , then  $n \leq a_1(a_n - 2n + 7 - \mu) + 2 - \epsilon$ .

**Proof.** From Lemma 3.7,

$$\begin{aligned} 2a_n - 4n + 8 - \mu &\geq |Z \setminus X| + |Z \setminus Y| \\ &\geq |Z| - |X \cap Y| = |Z| - |Y| + |Y \setminus X|. \end{aligned} \quad (3.26)$$

Because  $|Y| = n - 2$  and  $|Z| = a_n - n$ , it follows from Equation (3.26) that

$$\begin{aligned} |Y \setminus X| &\leq |Y| - |Z| + 2a_n - 4n + 8 - \mu \\ &= (n-2) - (a_n - n) + 2a_n - 4n + 8 - \mu \\ &= a_n - 2n + 6 - \mu. \end{aligned} \quad (3.27)$$

Note that  $y \in Y \setminus X$  if and only if  $y + a_2 = y + a_1 + 1 \in S$  and  $y + a_1 \notin S$ . Now, for all runs  $R_i$ , unless  $a_1, a_2 \in R_i$ , it holds that  $\min(R_i) - a_2 \in Y \setminus X$ , because by the definition of runs,  $\min(R_i) - a_2 + a_1 = \min(R_i) - 1 \notin S$ . Thus, from Equation (3.27),

$$t \leq |Y \setminus X| + 1 \leq a_n - 2n + 7 - \mu. \quad (3.28)$$

Next, because at most  $2 - \epsilon$  pairs of labels in  $S$  differ by  $a_1$  and  $2a_1 \notin S$ ,

$$\sum_{i=1}^t \max(|R_i| - a_1, 0) \leq 2 - \epsilon, \quad (3.29)$$

from which it follows that

$$|S| = \sum_{i=1}^t |R_i| \leq t \cdot a_1 + 2 - \epsilon. \quad (3.30)$$

Therefore, from Equations (3.27) and (3.28),  $|S| = n \leq a_1(a_n - 2n + 7 - \mu) + 2 - \epsilon$ .  $\square$

Next, we use the labels not in  $S \cup (T \setminus M)$  to find labels in  $S$ .

**Proposition 3.9.** If  $t \notin S$  and  $t \notin T \setminus M$ , then  $t + a_1 \in S$ . In addition, if  $t > 2a_1$  and  $t - a_1 \notin T \setminus M$ , then  $t - a_1 \in S$ .

**Proof.** First, if  $t + a_1 \notin S$ , then  $t \notin M$ , from which  $t \in T \setminus M$ . Next, if  $t > 2a_1$  and  $t - a_1 \notin S$ , then from  $S \sqcup T = [a_1, a_n]$ , it holds that  $t - a_1 \in T$ . Now, because  $t \notin S$ , it holds that  $t - a_1 \notin M$  by the definition of  $M$ . Therefore,  $t - a_1 \in T \setminus M$ , a contradiction.  $\square$

Next, we show that if  $\text{spum}(P_n) = 2n - 2$  and  $n \geq 13$ , then  $2a_1 \in L$ .

**Lemma 3.10.** If  $\text{spum}(P_n) = 2n - 2$  for  $n \geq 13$ , then  $2a_1 \in L$ .

**Proof.** Suppose otherwise. From Corollary 3.5,  $m = k = 0$ . By Lemma 3.6,  $a_1 \leq 2$ . First, if  $a_1 = 2$ , then  $|S \cap [2, 4]| = 2$ . Thus, because  $2a_1 = 4 \notin S$ , we have  $a_2 = 3$ . Then, because  $a_n \leq \text{spum}(P_n) = 2n - 2$ , by Lemma 3.8,  $n \leq 12$ . Thus,  $a_1 = 1$ .

Now, as  $m = 0$ , it holds that  $|S \cap [1, 2]| = 1$ , so  $2 \notin S$ . As  $k = 0$ , it holds that  $T \setminus M = \emptyset$ , so by setting  $t = 2$  in Proposition 3.9,  $a_2 = 3$ . Next, from Lemma 3.4,  $2n - 3 \leq a_n \leq 2n - 2$ . However, if  $a_n = 2n - 3$ , then  $N(2n - 3) = \emptyset$  as  $2 \notin S$ . Thus,  $a_n = 2n - 2$ .

Now, if  $2n - 3 \in S$ , then because  $2 \notin S$ , the vertices  $\{2n - 2, 1, 2n - 3\}$  are a connected component of  $G$ , so  $2n - 3 \notin S$ . Because  $k = 0$ , by setting  $t = 2n - 3$  in Proposition 3.9, it holds that  $2n - 4 \in S$ . Now, if  $4$  or  $2n - 5$  is in  $L$ , then  $\{1, 3, 4, 2n - 5, 2n - 4, 2n - 2\} \cap L$  is a connected component of  $G$ , so  $\{4, 2n - 5\} \subseteq T$ . Now, by setting  $t = 4$  (resp.  $2n - 5$ ) in Proposition 3.9, we have  $5 \in S$  (resp.  $2n - 6 \in S$ ). Because  $\{2, 4, 2n - 2, 2n - 4\} \subseteq T$ , it holds that  $|N(2n - 6)| = 1$ . However, the vertices  $2n - 2$  and  $2n - 4$  also have 1 neighbor. Thus,  $\{a_1, 2a_1\} \subseteq L$ .  $\square$

Now, from Lemma 3.10, we show that  $\text{spum}(P_n) \geq 2n - 1$ .

**Theorem 3.11.** If  $n \geq 8$ , then  $\text{spum}(P_n) \geq 2n - 1$ .

**Proof.** Suppose otherwise. First, from Theorem 3.1,  $\text{spum}(P_n) = 2n - 2$ . By computer search by Li [4],  $n \geq 13$ . Now, as  $m \geq -1$ , from Corollary 3.5,  $k \leq 1$ . By Lemma 3.6,  $a_1 \leq 3$ . However, from Lemma 3.10,  $2a_1 \in S$ . Thus,  $2n - 2 = \text{range}(L) \geq 3n - a_1 - 11$  by Corollary 3.3, which implies  $n \leq a_1 + 9 \leq 12$ .  $\square$

3.3. The inequality  $\text{spum}(P_n) \geq 2n$  for odd  $n \geq 9$

We now show that  $\text{spum}(P_n) \geq 2n$  for odd  $n \geq 9$ . In Section 3.4, we improve the bound to  $\text{spum}(P_n) \geq 2n + 1$  for odd  $n \geq 9$ . First, assign  $\varphi : [n] \rightarrow \{0, 1\}$  such that

$$\varphi(i) = \begin{cases} 1 & \text{if } a_i \geq 8 \text{ and } b_{i-1} = b_{i-2} = b_{i-3} = 2 \\ 0 & \text{otherwise.} \end{cases}$$

Next, we define for labels  $a_i$  in  $L$

$$\text{st}(a_i) = a_i - 2 \sum_{j=1}^{i-1} \varphi(j).$$

Note that  $\text{st}(a_i)$  is weakly increasing in  $i$ . Next, we define the set  $\text{st}(L)$

$$\text{st}(L) = \{\text{st}(a_i) \mid \varphi(i) = 0\}.$$

Note that each  $\text{st}(a_i)$  for  $i$  such that  $\varphi(i) = 0$  are distinct. The following proposition follows immediately from the definitions.

**Proposition 3.12.** If  $\text{st}(a_i) < \text{st}(a_j)$  are consecutive in  $\text{st}(L)$  for  $i \geq 2$ , then  $\text{st}(a_j) - \text{st}(a_i) = b_{j-1}$ .

**Proof.** If  $j = i + 1$ , it holds that  $\text{st}(a_j) - \text{st}(a_i) = a_{i+1} - a_i - 2\varphi(i) = b_i$ . Otherwise, if  $j \neq i + 1$ , then because  $\text{st}(a_\ell) \notin \text{st}(L)$  and  $\varphi(\ell) = 1$  for  $i \leq \ell - 1 \leq j$ , we have  $b_\ell = 2$  for  $i - 2 \leq \ell \leq j - 2$ . In addition, because  $\varphi(i + 1) = 1$  and  $\varphi(i) = 0$ , it follows that  $a_{i-3} \neq 2$ . Thus,  $\varphi(i - 1) = 0$ , and

$$\text{st}(a_j) - \text{st}(a_i) = b_{j-1} + 2 \cdot (j - i) - 2 \sum_{\ell=i-1}^{j-1} \varphi(\ell) = b_{j-1}. \quad \square \quad (3.31)$$

We now show that the number of pairs of labels that differ by 3 in  $L$  is at least that of  $\text{st}(L)$ .

**Table 1**  
Remaining possible  $\text{st}(L)$  in the proof of Lemma 3.14.

| $\text{st}(L)$                  | Why $P_n$ is not a sum graph of $L$               |
|---------------------------------|---------------------------------------------------|
| $\{1, 3, 5, 6, 7, 9, 11, 14\}$  | $\{1, 5, 6\}$ is a cycle.                         |
| $\{1, 3, 6, 7, 9, 11, 12\}$     | $N(2n-5) = \emptyset$                             |
| $\{1, 3, 4, 6, 8, 11, 12\}$     | $\{6, 8, 10\} \subseteq N(4)$ for $n \geq 9$ .    |
| $\{1, 3, 5, 6, 9, 11, 13, 14\}$ | $\{2n-1, 1, 5, 2n-5\}$ is a connected component.  |
| $\{1, 3, 5, 6, 8, 10, 13, 14\}$ | $\{1, 3, 2n-6\} \subseteq N(5)$                   |
| $\{1, 3, 5, 7, 8, 9, 12\}$      | $ N(n)  =  N(n-2)  =  N(2n-1)  = 1$ for odd $n$ . |
| $\{1, 3, 5, 7, 8, 11, 12\}$     | $ N(n)  =  N(n-2)  =  N(2n-1)  = 1$ for odd $n$ . |

**Proposition 3.13.** *It holds that  $|N_L(3)| \geq |N_{\text{st}(L)}(3)|$ .*

**Proof.** We show that if  $\text{st}(a_i) - 3 \in N_{\text{st}(L)}(3)$ , then  $a_i - 3 \in N_L(3)$ . If there exists some  $a_j$  such that  $\text{st}(a_i) = \text{st}(a_j) + 3$ . From Proposition 3.12, if  $[\text{st}(a_j) + 1, \text{st}(a_j) + 2] \cap \text{st}(L) = \emptyset$ , then  $b_{i-1} = 3$ . Thus,  $a_i - 3 \in N_L(3)$ . Next, if  $[\text{st}(a_j) + 1, \text{st}(a_j) + 2] \cap \text{st}(L) = \text{st}(a_j) + 1 = \text{st}(a_v)$  with  $\varphi(v) = 0$ , then  $b_{i-1} = 2$  and  $b_{v-1} = 1$  from Proposition 3.12. Now, because  $b_{v-1} = 1$ , it holds that if  $i \neq v + 1$ , then  $\varphi(v + 1) = 1$ . Thus,  $i = v + 1$ , so  $a_{v-1} = a_j - 3$ .

Now, suppose  $[\text{st}(a_j) + 1, \text{st}(a_j) + 2] \cap \text{st}(L) = \text{st}(a_j) + 2 = \text{st}(a_v)$  with  $\varphi(v) = 0$ . Then, from Proposition 3.12,  $b_{i-1} = 1$  and  $b_{v-1} = 2$ . If  $i = v + 1$ , then either  $a_{v-1} = a_i - 3$  or  $b_{i-2} = 2$  because  $\varphi(i - 1) = 1$ . Thus,  $a_{i-2} = a_j - 3$ .

Finally, suppose that  $[\text{st}(a_j) + 1, \text{st}(a_j) + 2] \subseteq \text{st}(L)$ . If  $\text{st}(a_j) + 1 = \text{st}(a_v)$  and in addition,  $\text{st}(a_j) + 2 = \text{st}(a_\ell)$  with  $\varphi(v) = \varphi(\ell) = 0$ , then from Proposition 3.12,  $b_{i-1} = b_{\ell-1} = b_{v-1} = 1$ . If  $i \neq \ell + 1$ , then  $\varphi(\ell + 1) = 1$  because  $b_{\ell-1} = 1$ . Thus,  $i = \ell + 1$ . Similarly,  $\ell = v + 1$ . Thus, because  $b_{i-1} = b_{\ell-1} = b_{v-1} = 1$ , we have  $a_i = a_{v-1} + 3$ .  $\square$

Next, we prove that if  $\text{spum}(P_n) = 2n - 1$  for odd  $n \geq 17$  and  $a_1 = 1$ , then  $2 \in S$ .

**Lemma 3.14.** *If  $\text{spum}(P_n) = 2n - 1$  for odd  $n \geq 17$  and  $a_1 = 1$ , then  $2 \in S$ .*

**Proof.** Suppose otherwise. Then  $m = 0$ . Now, from Corollary 3.5,  $k = |T \setminus M| \leq 1$ . If  $a_2 \geq 5$ , then  $\{2, 3, 4\} \subseteq T$ , which implies that  $\{2, 3\} \subseteq T \setminus M$ . It follows that  $a_2 \in \{3, 4\}$ . Suppose that  $a_2 = 4$ . Then  $2 \in T \setminus M$ , so  $k = 1$ . From Lemma 3.4,  $a_n \in \{2n - 2, 2n - 1\}$ . Because  $2 \in T$  and  $a_{n+1} = 2n - 1 + a_1 = 2n$ , if  $a_n = 2n - 2$ , then  $N(2n - 2) = \emptyset$ . Thus,  $a_n = 2n - 1$ . Now, as  $2 \notin S$  and  $|N(1)| \leq 2$ , if  $2n - 2 \in S$ , then the labels  $\{2n - 1, 1, 2n - 2\}$  form a connected component. Therefore,  $2n - 2 \in T$ . By setting  $t = 2n - 2$  in Proposition 3.9,  $2n - 3 \in S$ . However, if  $\{2, 3, 2n - 2\} \subseteq T$ , then  $N(2n - 3) = \emptyset$ . Thus,  $a_2 = 3$ .

Now, from Lemma 3.4,  $a_n \geq 2n - 3$ . Because  $2 \in T$  and  $a_{n+1} = \text{range}(L) + a_1 = 2n$ , if  $a_n = 2n - 2$ , then  $N(2n - 2) = \emptyset$ . Thus,  $a_n \in \{2n - 3, 2n - 1\}$ . Furthermore, if  $b_i \geq 3$ , then  $[a_i + 1, a_i + 2] \subseteq T$  and  $a_i + 1 \in T \setminus M$ . Therefore,  $b_i \geq 3$  for at most one value of  $i \in [1, n - 1]$ . In addition, because  $\sum_{i=1}^n b_i = \text{range}(L) = 2n - 1$ , it follows that  $c_1 = 2$ ,  $c_2 = n - 3$ , and  $c_3 = 1$ . Therefore, for  $n \geq 10$ , there exists a sequence  $b_i = b_{i+1} = b_{i+2} = 2$  with  $a_{i+2} \geq 8$ . Thus,  $|\text{st}(L)| \leq 9$ . From Proposition 3.13, we must have  $N_{\text{st}(L)}(3) \leq 2$ . In addition, if  $\{\max(\text{st}(L)) - 2, \max(\text{st}(L)) - 1\} \subseteq \text{st}(L)$ , then  $\{1, \max(L) - 2, \max(L) - 1\} \subseteq L$  is a connected component. Thus,  $\{\max(\text{st}(L)) - 2, \max(\text{st}(L)) - 1\} \not\subseteq \text{st}(L)$ . Now, Table 1 shows the exhaustive list of the possible  $\text{st}(L)$  found by computer search given the constraints

- (1)  $|\text{st}(L)| \leq 9$ ,
- (2)  $N_{\text{st}(L)}(3) \leq 2$ , and
- (3)  $\{\max(\text{st}(L)) - 2, \max(\text{st}(L)) - 1\} \not\subseteq \text{st}(L)$

and why each is not a sum graph labeling of  $P_n$ . Therefore,  $[1, 2] \subseteq S$ .  $\square$

Now, we prove the analog of Lemma 3.14 for  $a_1 \neq 1$ .

**Lemma 3.15.** *If  $\text{spum}(P_n) = 2n$  for odd  $n \geq 17$  and  $a_1 \neq 1$ , then  $2a_1 \in S$ .*

**Proof.** Suppose otherwise. Then from Corollary 3.5,  $k \leq 1$ . Thus,  $a_1 \in \{2, 3\}$ . First, suppose that  $a_1 = 3$ . Then,  $1 \leq |[a_n - 2, a_n - 1] \cup (T \setminus M)| \leq k$  by the definition of  $k$ . Thus,  $m = 0$  from Corollary 3.5, and so  $[3, 5] \subseteq S$ . Because  $a_n \geq 2n - 2$  from Lemma 3.4 and  $a_{n+1} = \text{range}(L) + 3 = 2n + 2$ , it follows that  $N(a_n) \cap \{3, 4\} \neq \emptyset$ , so  $\mu \geq 1$ . If  $a_n = 2n - 1$ , then  $\epsilon = 1$ , and from Lemma 3.8,  $n \leq 16$ . Thus,  $a_n = 2n - 2$ , so from Lemma 3.8,  $n \leq 14$ .

Next, suppose that  $a_1 = 2$ . First, if  $m = 1$ , then because  $\{3, 4\} \subseteq T$ ,  $\max(L) = 2n + 1$  and  $N(a_n) \neq \emptyset$ , from Lemma 3.4,  $a_n = 2n - 1$ . Now, because  $N(2n - 2) = \emptyset$ , then  $2n - 2 \in T \setminus M$ . Now, from Corollary 3.5,  $k = 0$ . Thus,  $m = 0$  and  $[2, 3] \subseteq S$ , so  $n \leq 14$  from Lemma 3.8.  $\square$

Next, from Lemmas 3.14 and 3.15, we arrive at the following corollary.

**Corollary 3.16.** *If  $\text{spum}(P_n) = 2n$  for odd  $n \geq 17$ , then  $2a_1 \in S$ .*

Now, we show that  $\text{spum}(P_n) \geq 2n$  for odd  $n \geq 9$ .

**Theorem 3.17.** *For odd  $n \geq 9$ , it holds that  $\text{spum}(P_n) \geq 2n$ .*

**Proof.** Suppose otherwise. First, from Theorem 3.11,  $\text{spum}(P_n) = 2n - 1$ . From computer search by Li [4],  $n \geq 17$ . Now, because  $m \geq -1$ , from Corollary 3.5,  $k \leq 2$ . Then, from Lemma 3.6,  $a_1 \leq 4$ . Furthermore, from Corollary 3.16,  $2a_1 \in S$ , and thus from Corollary 3.3,  $2n - 1 = \text{range}(L) \geq 3n - a_1 - 11$ , which implies that  $n \leq a_1 + 10 \leq 14$ .  $\square$

3.4. *The inequality  $\text{spum}(P_n) \geq 2n + 1$  for odd  $n \geq 9$*

Through a series of lemmas, we prove the claim that if  $\text{spum}(P_n) = 2n$  for odd  $n \geq 17$ , then  $2a_1 \in S$ . We first prove the claim for  $a_1 = 1$ . We begin by proving that if  $\text{spum}(P_n) = 2n$  for odd  $n \geq 17$  and  $a_1 = 1$ , then  $a_2 \leq 4$ . We then eliminate the possibility that  $a_2 \in \{3, 4\}$ . First, we show that  $a_2 \leq 4$ .

**Lemma 3.18.** *If  $\text{spum}(P_n) = 2n$  for odd  $n \geq 17$ , and  $a_1 = 1$ , then  $a_2 \leq 4$ .*

**Proof.** Suppose otherwise. Then  $m = 0$ . From Corollary 3.5,  $k = |T \setminus M| \leq 2$ . Now, if  $a_2 \geq 6$ , then  $\{2, 3, 4, 5\} \subseteq T$ , and thus,  $\{2, 3, 4\} \subseteq T \setminus M$ . Therefore,  $a_2 = 5$ .

As a result,  $\{2, 3\} \subseteq T \setminus M$ , so  $k = 2$ . From Lemma 3.4,  $a_n \geq 2n - 1$ . If  $a_n = 2n - 1$ , then  $N(2n - 1) = \emptyset$ , because  $2 \in T$  and  $a_{n+1} = \text{range}(L) + a_1 = 2n + 1$ . Thus,  $a_n = 2n$ . Now, if  $2n - 1 \in S$ , then because  $2 \notin S$  and  $|N(1)| \leq 2$ , the vertices with labels  $\{2n, 1, 2n - 1\}$  form a connected component. Therefore,  $2n - 1 \in T$ . By setting  $t = 2n - 1$  in Proposition 3.9,  $2n - 2 \in S$ . However, because  $\{2, 3, 2n - 1\} \subseteq T$ , it holds that  $N(2n - 2) = \emptyset$ .  $\square$

We now eliminate the possibility that  $a_2 = 4$ .

**Lemma 3.19.** *If  $\text{spum}(P_n) = 2n$  for odd  $n \geq 17$  and  $a_1 = 1$ , then  $a_2 \neq 4$ .*

**Proof.** Suppose otherwise. Then,  $2 \in T \setminus M$ , so  $k \geq 1$ . Now, from Lemma 3.4,  $a_n \geq 2n - 2$ . If  $a_n \in \{2n - 2, 2n - 1\}$ , then  $N(a_n) = \emptyset$ , because  $\{2, 3\} \subseteq T$  and  $a_{n+1} = 2n + 1$ . Thus,  $a_n = 2n$ .

First, if  $2n - 1 \in S$ , then because  $1 \in S$  and  $2 \notin S$  and  $|N(1)| \leq 2$ , the vertices with labels  $\{2n, 1, 2n - 1\}$  form a connected component. Therefore,  $2n - 1 \in T$ . Furthermore, if  $2n - 2 \in S$ , then because  $\{2, 3, 2n - 1\} \subseteq T$ , we have  $N(2n - 2) = \emptyset$ , so  $2n - 2 \in T$ . Then, from Corollary 3.5,  $k \leq 2$ , so  $\{2n - 2, 2\} = T \setminus M$ . Then, by setting  $t = 2n - 2$  in Proposition 3.9,  $2n - 3 \in S$ .

Now, if  $\{5, 2n - 4\} \cap L \neq \emptyset$ , then the vertices with labels in  $\{1, 4, 5, 2n - 4, 2n - 3, 2n\} \cap S$  form a connected component because  $|N(1)|, |N(4)| \leq 2$ . Thus, because  $n \geq 17$ , it must hold that  $\{5, 2n - 4\} \cap S = \emptyset$ . Now, setting  $t = 5$  (resp.  $t = 2n - 4$ ) in Proposition 3.9, we have  $6 \in S$  (resp.  $2n - 5 \in S$ ). But then, because  $\{2, 3, 5, 2n - 4, 2n - 2, 2n - 1\} \subseteq T$ , it holds that  $|N(2n - 5)| = 1$ . Therefore, the vertices with labels  $2n, 2n - 3$ , and  $2n - 5$  all have degree 1, a contradiction to  $G = P_n$ .  $\square$

We now eliminate the possibility that  $a_2 = 3$ .

**Lemma 3.20.** *If  $\text{spum}(P_n) = 2n$  for odd  $n \geq 17$  and  $a_1 = 1$ , then  $a_2 = 2$ .*

**Proof.** By Lemmas 3.18 and 3.19, it suffices to assume that  $a_2 = 3$  and reach a contradiction. Suppose  $a_2 = 3$ . Then from Lemma 3.4,  $a_n \geq 2n - 3$ . First, because  $|N(1)| \leq 2$ , then  $c_1 \leq 2$ . Then because  $\sum_{i=1}^n b_i = \text{range}(L) = 2n$ , either

- $c_2 = n$ ,
- $c_1 = c_3 = 2$ , and  $c_2 = n - 4$ ,
- $c_1 = 2$ ,  $c_4 = 1$ , and  $c_2 = n - 3$ , or
- $c_1 = 1$ ,  $c_2 = n - 2$ , and  $c_3 = 1$ .

First, if  $c_2 = n$ , then labels in  $L$  are odd, so  $G = I_n \neq P_n$ . Now, because  $2 \in T$ , we have  $b_n \neq 2$  and  $b_1 \neq 2$ , otherwise  $N(a_n) = \emptyset$  and  $a_2 = 2$ , respectively. Next, we show that if  $c_1 = 2$ ,  $c_4 = 1$ , and  $c_2 = n - 3$ , and  $b_n = 1$ , then  $b_{n-1} = 4$ . Suppose otherwise.

First,  $b_{n-1} \neq 1$ , otherwise  $\{2n, 1, 2n - 1\}$  is a connected component. In addition, if  $b_{n-1} = 2$ , then  $\{4, 2n - 3\} \cap L = \emptyset$ , otherwise  $\{1, 3, 4, 2n - 3, 2n - 2, 2n\} \cap S$  form a connected component, because  $2 \in T$ . In addition,  $2n - 4 \notin S$ , because otherwise  $\max(|N(2n - 4)|, |N(2n)|, |N(2n - 2)|) \leq 1$ . Thus,  $\{2n - 4, 2n - 3\} \subseteq T$ , and it follows that  $b_{n-2} = 4$  and  $a_{n-2} =$



**Table 2**  
Remaining possible  $\text{st}(L)$  in the proof of Lemma 3.20.

| $\text{st}(L)$                      | Why $P_n$ is not a sum graph of $L$ .                  |
|-------------------------------------|--------------------------------------------------------|
| $\{1, 3, 4, 5, 7, 9, 13\}$          | $\{1, 3, 5\} \subseteq N(4)$                           |
| $\{1, 3, 4, 6, 8, 9, 13\}$          | $ N(2n-6)  =  N(2n-4)  =  N(2n-3)  = 1$                |
| $\{1, 3, 4, 6, 8, 11\}$             | $\{1, 3, 2n-2\}$ is a connected component.             |
| $\{1, 3, 5, 6, 8, 10, 13\}$         | $\{1, 5, 2n-4\}$ is a connected component.             |
| $\{1, 3, 5, 7, 8, 11\}$             | $\{1, 2n-3\}$ is a connected component.                |
| $\{1, 3, 6, 7, 8, 10, 12, 15\}$     | $N(2n-4) = \emptyset$                                  |
| $\{1, 3, 6, 8, 10, 11, 12, 15\}$    | $ N(2n-4)  =  N(2n-3)  =  N(2n-2)  = 1$                |
| $\{1, 3, 5, 6, 7, 10, 12, 14, 17\}$ | $ N(2n-6)  =  N(2n-4)  =  N(2n-2)  = 1$                |
| $\{1, 3, 5, 7, 10, 11, 12, 15\}$    | $\{2n-9, 2n-7, 2n-4\} \subseteq N(5)$ for $n \geq 9$ . |
| $\{1, 3, 4, 6, 8, 12, 13\}$         | $\{1, 3, 2n\}$ is a connected component.               |
| $\{1, 3, 5, 6, 8, 10, 14, 15\}$     | $\{1, 3, 2n-4\} \subseteq N(5)$                        |
| $\{1, 3, 5, 7, 8, 10, 12, 16, 17\}$ | $ N(2n-6)  =  N(2n-4)  =  N(2n)  = 1$                  |
| $\{1, 3, 5, 7, 8, 10, 14, 15\}$     | $ N(2n-6)  =  N(2n-4)  =  N(2n)  = 1$                  |
| $\{1, 3, 5, 7, 8, 12, 13\}$         | $ N(3)  =  N(2n-4)  =  N(2n)  = 1$                     |
| $\{1, 3, 6, 7, 10, 12, 14, 15\}$    | $N(2n-4) = \emptyset$                                  |
| $\{1, 3, 6, 7, 9, 11, 14, 15\}$     | $\{1, 6, 2n-5, 2n\}$ is a connected component.         |
| $\{1, 3, 6, 8, 10, 11, 14, 15\}$    | $\{1, 2n-4, 2n\}$ is a connected component.            |
| $\{1, 3, 5, 6, 9, 11, 13, 16, 17\}$ | $\{5, 6, 2n-5\}$ is a cycle.                           |
| $\{1, 3, 5, 7, 10, 11, 14, 15\}$    | $ N(n)  =  N(n-2)  =  N(2n-1)  = 1$ for odd $n$ .      |

$2n-6$ . Because  $\max(|N(2n)|, |N(2n-2)|) \leq 1$ , it holds that  $|N(2n-6)| = 2$ , and  $\{6, 7\} \subseteq S$ . However,  $a_2 = 3$ ,  $\{6, 7\} \subseteq S$ ,  $c_3 = 0$ , and  $c_1 = 1$ . Therefore,  $b_{n-1} = 4$ .

Therefore, for  $n \geq 14$ , there exists  $i$  such that  $b_i = b_{i+1} = b_{i+2} = 2$  with  $a_{i+3} \geq 8$ . Thus,  $|\text{st}(L)| \leq 13$ . From Proposition 3.13,  $N_{\text{st}(L)}(3) \leq 2$ . In addition,  $\{\max(\text{st}(L)) - 2, \max(\text{st}(L)) - 1\} \not\subseteq \text{st}(L)$ , otherwise  $\{\max(L) - 2, \max(L) - 1, 1\} \subseteq L$  is a connected component. Now, Table 2 shows the exhaustive list of the possible  $\text{st}(L)$  found by computer search given the constraints

- (1)  $|\text{st}(L)| \leq 13$ ,
- (2)  $N_{\text{st}(L)}(3) \leq 2$ , and
- (3)  $\{\max(\text{st}(L)) - 2, \max(\text{st}(L)) - 1\} \not\subseteq \text{st}(L)$

and why each is not a sum graph labeling of  $P_n$ . Thus,  $a_2 \neq 3$  as sought.  $\square$

Now, the following corollary follows from Lemmas 3.18, 3.19, and 3.20.

**Corollary 3.21.** *If  $\text{spum}(P_n) = 2n$  for odd  $n \geq 17$  and  $a_1 = 1$ , then  $2 \in S$ .*

Next, we show the analog of Corollary 3.21 for  $a_1 = 2$ .

**Lemma 3.22.** *If  $\text{spum}(P_n) = 2n$  for odd  $n \geq 17$  and  $a_1 = 2$ , then  $4 \in S$ .*

**Proof.** Suppose otherwise. Because  $m \geq 0$ , from Corollary 3.5,  $k \leq 2$ . If  $a_2 \geq 6$ , then  $m = 1$  and  $\{2, 3\} \subseteq T \setminus M$ . Then, from Corollary 3.5,  $\text{range}(L) \geq 2n + 1$ . Thus, because  $2a_1 = 4 \in T$ , it holds that  $a_2 \in \{3, 5\}$ . First, if  $a_2 = 3$ , then because  $a_n \leq \text{range}(L)$ , from Lemma 3.8,  $n \leq 16$ . Thus,  $a_2 = 5$ .

Because  $a_2 = 5$ , it follows that  $m = 1$ , and because  $3 \in T \setminus M$ , from Corollary 3.5,  $k = 1$ . From Lemma 3.4,  $2n - 1 \leq a_n \leq 2n$ . Because  $3 \in T$ , if  $2n - 1 \in S$ , then  $N(2n - 1) = \emptyset$ . Thus, because  $\{2n - 1, 2\} \subseteq T \setminus M$ , it follows that  $k \geq 2$ . Finally, from Corollary 3.5,  $\text{range}(L) \geq 2n + 1$ . Thus,  $2a_1 \in S$ .  $\square$

Next, we show the analog of Corollary 3.21 for  $a_1 = 3$ .

**Lemma 3.23.** *If  $\text{spum}(P_n) = 2n$  for odd  $n \geq 17$ , and  $a_1 = 3$ , then  $6 \in S$ .*

**Proof.** Suppose otherwise. Then, as  $m \geq 0$ , it follows from Corollary 3.5 that  $k \leq 2$ . Next, from Lemma 3.4,  $2n - 3 \leq a_n \leq 2n$ . But, because  $N(a_n) \neq \emptyset$  and  $a_{n+1} = a_1 + \text{range}(L) = 2n + 3$ , it holds that  $a_n \neq 2n - 3$ . Next, because  $6 \in T$  and  $a_{n+1} = 2n + 3$ ,

if  $a_n = 2n - 2$ , then from Lemma 3.4,  $\{2n - 3\} = T \setminus M$ , and  $m = 0$ . Thus,  $\{3, 5\} \subseteq S$ . In addition, by setting  $t = 6$  (resp.  $t = 2n - 3, 2n - 1$ ) in Proposition 3.9,  $9 \in S$  (resp.  $\{2n - 4, 2n - 6\} \subseteq S$ ). Because  $N(2n - 4) \neq \emptyset$ , it holds that  $7 \in S$ . Thus,  $\{3, 5, 2n - 6\} \subseteq N(4)$ , so  $a_n \neq 2n - 2$ .

Now, if  $a_n = 2n - 1$  then  $4 \in S$ , otherwise  $N(2n - 1) = \emptyset$ . In addition,  $\{2n - 3\} = T \setminus M$ , otherwise  $N(2n - 3) = \emptyset$ . Because  $N(2n - 2) \neq \emptyset$ , if  $2n - 2 \in S$ , then  $5 \in S$ . Therefore, if  $\{9, 2n - 6\} \cap S \neq \emptyset$ , then  $\{4, 5, 2n - 6, 2n - 2, 2n - 1\}$  is a connected component. Thus,  $\{9, 2n - 6\} \subseteq T$ , and  $\{6, 2n - 6, 2n - 3\} \subseteq T \setminus M$ . Because  $k \leq 2$  from Corollary 3.5, we have that  $2n - 2 \in T$ . Then from Lemma 3.4,  $\{2n - 3, 2n - 2\} = T \setminus M$ , and  $m = 0$ . Therefore,  $\{3, 5\} \subseteq S$ , and  $\{6, 2n - 5\} \cap T \setminus M = \emptyset$ . Thus, by setting  $t = 6$  (resp.  $t = 2n - 2$ ) in Proposition 3.9,  $9 \in S$  (resp.  $2n - 5 \in S$ ) which implies  $\{5, 2n - 5, 2n - 1\} \subseteq N(4)$ . Therefore,  $a_n \neq 2n - 1$ .

Therefore,  $a_n = 2n$ . If  $a_{n-1} = 2n - 1$  then  $4 \in S$ , as otherwise  $N(2n - 1) = \emptyset$ . Thus,  $\mu = 2$ , and  $\epsilon = 1$ . Now, because  $n \geq 17$  and from Lemma 3.8,  $a_{n-1} \leq 2n - 2$ . If  $a_{n-1} \leq 2n - 3$ , then from Lemma 3.4,  $\{2n - 2, 2n - 1\} = T \setminus M$ . Next, because  $m = 0$ , it follows that  $\{3, 5\} \subseteq S$ . Thus, by setting  $t = 6$  (resp.  $t = 2n - 2, 2n - 1$ ) in Proposition 3.9,  $9 \in S$  (resp.  $\{2n - 5, 2n - 4\} \subseteq S$ ). Now, if  $7 \in S$  (resp.  $8 \in S$ ), then  $\{3, 5, 2n - 4\} \subseteq N(4)$  (resp.  $\{3, 4, 2n - 5\} \subseteq N(5)$ ). Therefore,  $\{7, 8\} \subseteq T$ , which implies  $\{2n - 4, 4, 5, 2n - 5\}$  is a connected component, so  $a_{n-1} = 2n - 2$ . Now, because  $N(2n - 2) \neq \emptyset$ , it holds that  $5 \in S$ . Next, if  $2n - 4 \in S$ , because  $|N(2n - 2)| = |N(2n)| = 1$ , it follows that  $|N(2n - 4)| = 2$ , which implies  $\{4, 7\} \subseteq S$ . Now, because  $N(4) = \{3, 2n - 4\}$ , it holds that  $9 \in T$ , and from Lemma 3.4,  $\{6, 2n - 1\} = T \setminus M$ . Thus, because  $9 \notin T \setminus M$ , by setting  $t = 9$  in Proposition 3.9,  $12 \in S$ . Now, because  $|N(2n - 2)| = |N(2n)| = 1$ , it holds that  $\{3, 4, 5, 7, 2n - 4, 2n - 2, 2n\}$  is a connected component. Thus,  $2n - 4 \in T$ , and from Lemma 3.4,  $\{2n - 4, 2n - 1\} = T \setminus M$ , and  $m = 0$ , so  $\{3, 5\} \subseteq S$ . Thus, because  $6, 2n - 7 \notin T \setminus M$ , by setting  $t = 6$  (resp.  $t = 2n - 4$ ) in Proposition 3.9,  $9 \in S$  (resp.  $2n - 7 \in S$ ). Therefore,  $\{4, 2n - 7, 2n - 2\} \subseteq N(5)$ . Thus, because  $\Delta = 2$ , it holds that  $2a_1 \in S$ .  $\square$

Now, we show the analog of Corollary 3.21 for  $a_1 \geq 4$ .

**Lemma 3.24.** *If  $\text{spum}(P_n) = 2n$  for odd  $n \geq 17$  and  $a_1 \geq 4$ , then  $2a_1 \in S$ .*

**Proof.** Suppose otherwise. Then from Corollary 3.5 and Lemma 3.6,  $a_1 = 4$ ,  $k = 2$  and  $m = 0$ . Therefore,  $\{4, 7\} \subseteq S$ . Because  $|[a_n - 3, a_n - 1] \cup (T \setminus M)| \geq 2$  by the definition of  $k$ , it holds that  $T \setminus M \subseteq [a_n - 3, a_n - 1]$ . Thus, by setting  $t = 8 \notin T \setminus M$  in Proposition 3.9,  $12 \in S$ . In addition,  $a_{n+1} = a_1 + \text{spum}(P_n) = 2n + 4$ , and from Lemma 3.4,  $a_n \in \{2n - 1, 2n\}$ .

If  $a_n = 2n - 1$ , then  $\{7, 2n - 1\} = N(5)$ . If  $2n - 3 \in S$ , then because  $12 \in S$ , it follows that  $\{2n - 1, 5, 7, 2n - 3\}$  is a connected component. If  $2n - 4 \in S$ , then because  $8 \in S$ , it follows that  $N(2n - 4) = \emptyset$ . Thus, from Corollary 3.5,  $T \setminus M = \{2n - 4, 2n - 3\}$ . Now, by setting  $t = 2n + 2$  in Proposition 3.9,  $2n - 2 \in S$ . If  $13 \in S$ , then  $\{2n, 5, 7, 6, 2n - 1\}$  is a connected component. Therefore,  $13 \in T$ . Furthermore, because  $4 \notin N(5)$ , it follows that  $9 \in T \setminus M$ , so  $a_n = 2n$ .

Now, by the definition of  $k$ ,  $|[a_n - 3, a_n - 1] \cap S| = 1$ . If  $2n - 3 \in S$ , then  $9 \notin S$  as otherwise,  $\{2n, 4, 5, 7, 2n - 3\}$  is a connected component. Thus,  $9 \in T$ , and by setting  $t = 9 \notin T \setminus M$  in Proposition 3.9,  $13 \in S$ . Then,  $\{5, 6, 2n - 3\} \subseteq N(7)$ , so  $2n - 3 \notin S$ . If  $2n - 2 \in S$ , then by setting  $t = 2n - 1 \notin T \setminus M$  in Proposition 3.9,  $a_n - 5 \in S$ . Thus,  $\{7, 2n - 5\} = N(5)$ . Because  $4 \in S$ , but  $4 \notin N(5)$ , it holds that  $9 \in T$ . But then because  $\{9, 2n - 1\} \subseteq T$ , it holds that  $|N(2n - 5)| = |N(2n - 2)| = |N(2n)| = 1$ , so  $2n - 2 \notin S$ . Therefore,  $2n - 5 \in S$ , and so  $|N(5)| = 2$  implies that  $a_n - 6 \in T \setminus M$ , a contradiction with  $T \setminus M \subseteq [a_n - 3, a_n - 1]$ . Thus,  $2a_1 \in S$ .  $\square$

Now, from Corollary 3.21 and Lemmas 3.22, 3.23, and 3.24, it follows that if  $\text{spum}(P_n) = 2n$ , then  $2a_1 \in S$ .

**Corollary 3.25.** *If  $\text{spum}(P_n) = 2n$  for odd  $n \geq 17$ , then  $2a_1 \in S$ .*

Finally, from Corollary 3.25, we evaluate  $\text{spum}(P_n)$  for  $n \geq 3$ .

**Theorem 3.26.** *For  $n \geq 3$ , it holds that*

$$\text{spum}(P_n) = \begin{cases} 2n - 3 & \text{if } 3 \leq n \leq 6 \\ 2n - 2 & \text{if } n = 7 \\ 2n - 1 & \text{if } n \geq 8 \text{ is even} \\ 2n + 1 & \text{if } n \geq 9 \text{ is odd.} \end{cases}$$

**Proof.** The statement is true for  $n \leq 15$  from computer search by Li [4]. From Theorem 3.17,  $\text{spum}(P_n) \geq 2n$  for odd  $n \geq 9$ . Now, assume for the sake of contradiction that  $\text{range}(L) = 2n$  for odd  $n \geq 17$ . First, because  $m \geq -1$ , from Corollary 3.5,  $k \leq 3$ . Then from Lemma 3.6,  $a_1 \leq 5$ . Furthermore, from Corollary 3.25,  $2a_1 \in S$ . Thus, from Corollary 3.3,  $2n = \text{range}(L) \geq 3n - a_1 - 11$ , which implies that  $n \leq a_1 + 11 \leq 16$ . Lastly, from Theorem 3.1,  $\text{spum}(P_n) \leq 2n + 1$ .  $\square$

#### 4. The integral sum-diameter of paths

In this section, we prove Theorem 4.32, from which Theorem 4.21 follows. Currently, the best known lower bound on  $\text{isd}(P_n)$  is by Singla, Tiwari, and Tripathi [5], and the best known upper bound is by Li [4].

**Theorem 4.1** ([5, Theorem 2.2] and [4, Proposition 9.6]). For  $n \geq 3$ , it holds that

$$2n - 5 \leq \text{isd}(P_n) \leq \begin{cases} 2n - 2 & \text{if } n \text{ is even} \\ 2n - 3 & \text{if } n \text{ is odd} \end{cases}.$$

For the rest of this section, let  $G \cup I_t$  be an integral sum graph of  $L$  that satisfies  $\text{range}(L) = \text{isd}(G)$ . Let  $L = \{a_1, a_2, \dots, a_{n+t}\}$  with  $a_1 < \dots < a_{r-1} < 0 < a_r < \dots < a_{n+t}$ , and let  $S \subseteq L$  be the labels of the vertices of  $G$ . Additionally, because  $-L$  is still an integral sum graph labeling of  $G \cup I_t$ , we assume without loss of generality that  $a_r \leq -a_{r-1}$ . Now, define  $b_i = a_{i+1} - a_i$  for  $1 \leq i \leq n+t-1$ , and  $c_\ell$  be the number of  $1 \leq i \leq n+t-1$  such that  $b_i = \ell$ . For each  $1 \leq i \leq n+t$ , say that  $a_i$  and  $a_{i+1}$  are consecutive. Finally, denote  $N_L(\ell) = N(\ell)$  unless otherwise specified.

Next, we borrow notations from [5], and let

$$\begin{aligned} S_1 &= \{a_1, a_2, \dots, a_{r-1}\}, \\ S_2 &= \{a_r, a_{r+2}, \dots, a_{n+t}\}, \\ S_3 &= S_1 + a_r, \text{ and} \\ S_4 &= S_2 - a_r. \end{aligned}$$

In addition, for this section, let  $T = [a_1, a_{n+t}] \setminus L$  and  $M = S_3 \cup S_4$ . Finally, let  $\eta = 1$  if  $a_{r-1} = -a_r$ , and  $\eta = 0$  otherwise. Similarly, let  $\xi = 1$  if  $2a_r \in L$ , and  $\xi = 0$  otherwise.

##### 4.1. The inequality $\text{isd}(P_n) \geq 2n - 4$ for $n \geq 8$

First, we set a lower bound on  $|T \setminus M|$ .

**Lemma 4.2.** It holds that  $|([a_{r-1} - a_r + 1, -a_r - 1] \cup [a_r, 2a_r - 1]) \setminus L| \leq |T \setminus M|$ .

**Proof.** First, because  $S_3 = S_1 + a_r = M \cap \mathbb{Z}_{\leq 0}$ ,

$$|[a_{r-1} - a_r + 1, -a_r - 1] \setminus S_1| = |[a_{r-1} + 1, -1] \setminus S_3| = |[a_{r-1} + 1, -1] \setminus M|. \quad (4.1)$$

Next, because  $S_4 = S_2 - a_r = M \cap \mathbb{Z}_{\geq 0}$ ,

$$|[a_r, 2a_r - 1] \setminus S_2| = |[0, a_{r-1} - 1] \setminus S_4| = |[0, a_r] \setminus M|. \quad (4.2)$$

Now, from Equations (4.1) and (4.2), it holds that

$$|([a_{r-1} - a_r + 1, -a_r - 1] \cup [a_r, 2a_r - 1]) \setminus L| \leq |[-a_{r-1} + 1, a_r - 1] \setminus M|. \quad (4.3)$$

Now, by the definitions of  $a_{r-1}$  and  $a_r$ ,

$$[a_{r-1} + 1, a_r - 1] \setminus M \subseteq T \setminus M. \quad (4.4)$$

It follows from Equation (4.3) that the lemma holds.  $\square$

Now, we generalize Theorem 2.2 of [5].

**Proposition 4.3.** It holds that  $|S_1 \cap S_3| + |S_2 \cap S_4| = N(a_r) + \xi$ .

**Proof.** First, because  $a_r + a_{r-1} \leq 0$ ,

$$N(a_r) \cap S_1 = S_1 \cap S_3. \quad (4.5)$$

Similarly, it holds that

$$N(a_r) \cap S_2 = (S_2 \cap S_4) \setminus \{a_r\}. \quad (4.6)$$

Finally, as  $a_r \in S_2 \cap S_4$  if and only if  $2a_r \in S_2$ , it holds that

$$|(S_2 \cap S_4) \setminus \{a_r\}| = |S_2 \cap S_4| - \xi. \quad (4.7)$$

Now, because  $S_1 \sqcup S_2 = L$ , from Equations (4.5), (4.6), and (4.7),

$$|N(a_r)| = |S_1 \cap S_3| + |S_2 \cap S_4| - \xi. \quad \square \quad (4.8)$$

Next, we strengthen Proposition 4.3.

**Proposition 4.4.** *It holds that  $\sum_{i=1}^4 |S_i| - |\bigcup_{i=1}^4 S_i| = |N(a_r)| + \xi + \eta$ .*

**Proof.** First,  $S_3 \cap S_4 = \{0\}$  if and only if  $a_{r-1} = -a_r$ . Thus, it holds that

$$|S_3 \cap S_4| = \eta. \quad (4.9)$$

Next, because  $a_{r-1} + a_r \leq 0$ , it follows from the definitions of  $S_1, S_2, S_3, S_4$  that

$$S_1 \cap S_4 = \emptyset, S_2 \cap S_3 = \emptyset, S_1 \cap S_2 = \emptyset. \quad (4.10)$$

Thus, from Proposition 4.3 and Equations (4.9) and (4.10),

$$\sum_{i=1}^4 |S_i| - \left| \bigcup_{i=1}^4 S_i \right| = |S_1 \cap S_3| + |S_2 \cap S_4| + |S_3 \cap S_4| = |N(a_r)| + \xi + \eta. \quad \square \quad (4.11)$$

Now, we use Proposition 4.4 to refine Theorem 2.2 of [5] to  $\text{isd}(P_n)$ .

**Lemma 4.5.** *It holds that  $\text{isd}(P_n) = 2(n+t) + |T \setminus M| - |N(a_r)| - \xi - \eta - 1$ .*

**Proof.** First, because  $\bigcup_{i=1}^4 S_i = (S_1 \sqcup S_2) \cup (S_3 \cup S_4) = L \cup M$ , it holds that

$$[a_1, a_{n+t}] \setminus \bigcup_{i=1}^4 S_i = [a_1, a_{n+t}] \setminus (L \cup M) = T \setminus M. \quad (4.12)$$

Now, because  $\bigcup_{i=1}^4 S_i \subseteq [a_1, a_{n+t}]$ , from Equation (4.12),

$$|[a_1, a_{n+t}]| - \left| \bigcup_{i=1}^4 S_i \right| = |T \setminus M|. \quad (4.13)$$

Therefore, from Equation (4.13) and Proposition 4.4, it holds that

$$\begin{aligned} \text{isd}(P_n) &= a_{n+t} - a_1 = |[a_1, a_{n+t}]| - 1 = \left| \bigcup_{i=1}^4 S_i \right| + |T \setminus M| - 1 \\ &= \sum_{i=1}^4 |S_i| + |T \setminus M| - |N(a_r)| - \xi - \eta - 1 \\ &= 2(n+t) + |T \setminus M| - |N(a_r)| - \xi - \eta - 1. \quad \square \end{aligned} \quad (4.14)$$

If  $\text{isd}(P_n) = 2n - 5$ , then Lemma 4.5 results in the following corollary.

**Corollary 4.6.** *If  $\text{isd}(P_n) = 2n - 5$ , then  $t = 0$ ,  $T \subseteq M$ ,  $|N(a_r)| = 2$ , and  $\xi = \eta = 1$ .*

Next, we show that if  $[a, b]$  and  $[c, d]$  are disjoint intervals that share sufficiently many elements with  $L$ , then another interval is disjoint from  $L$ . Let  $\tau = |([a, b] \sqcup [c, d]) \setminus L|$ .

**Lemma 4.7.** *If  $[a, b]$  and  $[c, d]$  are disjoint intervals such that  $\min(b-a, d-c) \geq 2 + \tau$ , then*

$$([2 + \tau + a - d, -2 - \tau + b - c] \cup [2 + \tau + c - b, d - a - 2 - \tau]) \cap L = \emptyset.$$

**Proof.** We first show that  $[2 + \tau + a - d, -2 - \tau + b - c] \cap L = \emptyset$ . Suppose otherwise, and let  $\ell = 2 + \tau + a - d + k \in [2 + \tau + a - d, -2 - \tau + b - c] \cap L$ . Because  $\ell \leq -2 - \tau + b - c$ ,

$$0 \leq k \leq d - c + b - a - 4 - 2\tau. \quad (4.15)$$

Consider  $([c, d] + \ell) \cap [a, b] = [c + \ell, d + \ell] \cap [a, b]$ . From Equation (4.15),

$$c + \ell = 2 + \tau + a + c - d + k \leq b - 2 - \tau, \text{ and} \quad (4.16)$$

$$d + \ell = 2 + \tau + a + k \geq a + 2 + \tau. \quad (4.17)$$

Thus, from Equations (4.16) and (4.17), either  $[b - 2 - \tau, b] \subseteq ([c, d] + \ell)$ , or  $[a, a + 2 + \tau] \subseteq [c, d] + \ell$ , or  $[c, d] + \ell \subseteq [a, b]$ . Then

$$|[a, b] \cap ([c, d] + \ell)| \geq 3 + \tau. \quad (4.18)$$

Now, because  $|([a, b] \cup [c, d]) \setminus L| \leq \tau$ , and if  $\{t, t + \ell\} \subseteq L$  then  $t \in N(\ell)$  from Equation (4.18), it holds that  $N(\ell) \geq 3$ , a contradiction as  $G = P_n$ . Lastly, swapping  $[a, b]$  and  $[c, d]$  gives  $[2 + \tau + a - d, -2 - \tau + b - c] \cap L = \emptyset$ .  $\square$

We now show that if  $T \setminus M$ , and  $\{-2a_r, -a_r, a_r, 2a_r\} \subseteq L$ , then  $a_r = 1$ .

**Lemma 4.8.** *If  $T \subseteq M$ ,  $\xi = \eta = 1$ ,  $t \leq 1$ ,  $-2a_r \in L$ , and  $n \geq 9$ , then  $a_r = 1$ .*

**Proof.** Because  $P_n$  is a integral sum graph of  $-L$ , assume without loss of generality that  $a_{n+t} \geq -a_1$ . Now, because  $T \subseteq M$ ,  $-2a_r \in L$ , and from Lemma 4.2,

$$[-2a_r, -a_r] \cup [a_r, 2a_r] \subseteq L. \quad (4.19)$$

Next, because  $t \leq 1$ , we have  $\{2a_r - 1, -2a_r + 1\} \cap S \geq 1$ . Thus, because  $N(2a_r - 1) \neq \emptyset$  or  $N(-2a_r + 1) \neq \emptyset$ , and  $a_{n+t} \geq -a_1$ , from Equation (4.19),  $a_n \geq 3a_r - 1$ .

Now, because  $a_n \geq 3a_r - 1 \geq 4a_r - 2$  for  $a_r \geq 3$ , by applying Lemma 4.7 on intervals  $[-2a_r, -a_r]$  and  $[a_r, 2a_r]$ , it holds that

$$[2a_r + 2, 4a_r - 2] \subseteq T. \quad (4.20)$$

Thus, if  $a_r \geq 4$ , then  $\{2a_r + 2, 3a_r + 2\} \subseteq T$ , so  $2a_r \in T \setminus M$ . Thus,  $a_r \leq 3$ .

Now, if  $a_r = 3$ , because  $a_n \geq 3a_r - 1 = 8$ , and setting  $a_r = 3$  in Equation (4.20) implies  $[8, 10] \subseteq T$ , we have  $a_n \geq 11$ . Thus, because  $\{7, 8\} \cap T \setminus M = \emptyset$ , and  $[8, 10] \subseteq T$ , we have  $\{7, 11\} \subseteq L$ . But, by Equation (4.19),  $\{-3, -4, 4\} \subseteq N(7)$ . Thus,  $a_r \neq 3$ .

Finally, if  $a_r = 2$ , by applying Lemma 4.7 on intervals  $[-4, -2]$  and  $[2, 4]$ , it follows that  $\{-6, 6\} \cap L = \emptyset$ . Now, because  $n \geq 9$ , and  $a_n \geq -a_1$ , it follows that  $a_n \geq 7$ . Thus,  $6 \in T \subseteq M$ , which implies  $8 \in L$ . Now, if  $7 \in L$  then  $N(-4) = \{2, 7, 8\}$ . Thus,  $7 \in T$ . However, if  $5 \in L$ , then  $\{-3, -2, 3\} \subseteq N(5)$ . Thus,  $5 \in T \setminus M$ . Therefore,  $a_r = 1$ .  $\square$

We now show that if  $\text{isd}(P_n) = 2n - 5$  and  $n \geq 9$ , then  $a_r = 1$ .

**Corollary 4.9.** *If  $\text{isd}(P_n) = 2n - 5$  for  $n \geq 9$ , then  $a_r = 1$ .*

**Proof.** Because  $P_n$  is a integral sum graph of  $-L$ , by applying Corollary 4.6 to  $L$  and  $-L$ , we have  $\xi = \eta = 1$ ,  $T \subseteq M$ ,  $t = 0$ , and  $-2a_r \in L$ . Thus, by Lemma 4.8,  $a_r = 1$ .  $\square$

Now, we bound  $\text{isd}(P_n)$  from below assuming  $a_r = 1$ ,  $a_{r-1} + a_r = 0$ , and  $2a_r \in L$ .

**Lemma 4.10.** *If  $T \subseteq M$  and  $a_r = \xi = \eta = 1$ , then  $\text{range}(L) \geq 3n - 12$ .*

**Proof.** Because  $a_r = \xi = 1$ , it follows that  $\{1, 2\} \subseteq L$ . Now, if  $i \neq r$ , and  $b_i = 1$ , then  $a_i \in N(1)$ . Thus,  $c_1 \leq 3$  because  $|N(1)| \leq 2$ . Likewise, if  $b_i = 2$  and  $i \neq r + 1$ , then  $a_i \in N(2)$ . Thus,  $c_2 \leq 3$ , because  $|N(2)| \leq 2$ . Therefore,

$$\text{range}(L) = \sum_{i=1}^{n+t-1} b_i \geq c_1 \cdot 1 + c_2 \cdot 2 + (n + t - 1 - c_1 - c_2) \cdot 3 \geq 3n - 12. \quad \square \quad (4.21)$$

We now use Lemmas 4.9 and 4.10 to show that  $\text{isd}(P_n) \geq 2n - 4$  when  $n \geq 8$ .

**Lemma 4.11.** *If  $n \geq 8$ , then  $\text{isd}(P_n) \geq 2n - 4$ .*

**Proof.** Suppose otherwise. Then  $\text{range}(L) = 2n - 5$  by Theorem 4.1. First,  $\text{isd}(P_n) \geq 2n - 4$  by exhaustive search for  $n = 8$ . Next, if  $n \geq 9$ , then  $a_r = \xi = \eta = 1$  from Corollary 4.6 and Corollary 4.9. Thus,  $\text{range}(L) \geq 3n - 12 > 2n - 5$  from Lemma 4.10.  $\square$

4.2. The inequality  $\text{isd}(P_n) \geq 2n - 3$  for  $n \geq 8$

Now, we state a corollary of Lemma 4.5 if  $\text{isd}(P_n) = 2n - 4$ .

**Corollary 4.12.** If  $\text{isd}(P_n) = 2n - 4$  and  $n \geq 9$ , then  $t = 0$ , and  $|T \setminus M| = \xi + \eta - 1$ .

**Proof.** From Lemma 4.5, if  $\text{isd}(P_n) = 2n - 4$ , then  $2t + |T \setminus M| \geq |N(a_r)| + \xi + \eta - 3$ . Because  $\max(\xi, \eta) \leq 1$ , and  $|N(a_r)| \leq 2$ , it holds that  $t = 0$ . Now, if  $|N(a_r)| \leq 1$ , then  $T \subseteq M$  and  $\xi = \eta = 1$ . Thus, from Lemma 4.8 and Lemma 4.10,  $a_r = 1$  and  $\text{range}(L) \geq 3n - 12 > 2n - 4$ , because  $n \geq 9$ . Thus,  $|N(a_r)| = 2$ , and  $|T \setminus M| = \xi + \eta - 1$  from Lemma 4.5.  $\square$

Now, let  $k = |([-2a_r, -a_r] \cup [a_r, 2a_r]) \cap T|$ . In addition, let

$$J_1 = ([a_1, -2a_r - 1] \cap L) + 2a_r,$$

and

$$J_2 = ([2a_r + 1, a_{n+t}] \cap L) - 2a_r.$$

We now prove the analog of Corollary 3.3 for  $\text{isd}(P_n)$ .

**Lemma 4.13.** If  $-2a_r \in L$  or  $2a_r \in L$ , then  $\text{range}(L) \geq 3n + 3t + k - 2a_r - 11$ .

**Proof.** First,  $J_1 \cap J_2 = \emptyset$ . Additionally, because  $S \sqcup T = [a_1, a_n]$ ,

$$\begin{aligned} |J_1 \cup J_2| &= |([a_1, -2a_r - 1] \cup [2a_r + 1, a_{n+t}]) \cap L| \\ &= n + t - |([-2a_r, -a_r] \cup [a_r, 2a_r]) \cap L| \\ &= n + t - 2(a_r + 1) + k. \end{aligned} \quad (4.22)$$

Now, because  $(S_3 \cap S_1) - a_r \subseteq N(a_r)$ , and  $S_4 \cap S_2 \subseteq N(a_r) \cup \{a_r\}$ , but  $(S_1 - a_r) \cap S_2 = \emptyset$ ,

$$|(S_3 \cap S_1)| + |S_4 \cap S_2| \leq |N(a_r) \cup \{a_r\}| \leq 3. \quad (4.23)$$

First, if  $2a_r \in L$ , then  $(J_1 \cap S_1) - 2a_r \subseteq N(2a_r) \setminus \{-a_r\}$  and  $J_2 \cap S_2 \subseteq N(2a_r) \cup \{2a_r\}$ . Next, because  $(S_1 - 2a_r) \cap S_2 \neq \emptyset$  and  $-a_r \in N(2a_r)$  if and only if  $\eta = 1$ ,

$$|(J_1 \cap S_1)| + |J_2 \cap S_2| \leq |N(a_r) \cup \{2a_r\} \setminus \{-a_r\}| \leq 3 - \eta. \quad (4.24)$$

Equivalently, if  $-2a_r \in L$ , then  $J_1 \cap S_1 \subseteq N(-2a_r) \cup \{-2a_r\}$  and  $J_2 \cap S_2 \subseteq N(-2a_r) \setminus \{a_r\}$ , but  $a_r \in N(-2a_r)$  if and only if  $\eta = 1$ . Thus, Equation (4.24) holds.

Now, because  $S_1 \sqcup S_2 \sqcup T = [a_1, a_{n+t}]$ , and  $S_3 \cap S_4 \subseteq \{0\}$ , from Equation (4.23),

$$\begin{aligned} |T \cap (S_3 \cup S_4)| &= |S_3 \cup S_4| - |S \cap (S_3 \cup S_4)| \\ &\geq (n + t - \eta) - |(S_3 \cap S_1)| - |(S_4 \cap S_2)| \\ &\geq (n + t - \eta) - 3. \end{aligned} \quad (4.25)$$

Likewise, because  $S_1 \sqcup S_2 \sqcup T = [a_1, a_{n+t}]$  and  $J_1 \cap J_2 = \emptyset$ , from Equations (4.22) and (4.24),

$$\begin{aligned} |T \cap (J_1 \cup J_2)| &= |J_1 \cup J_2| - |S \cap (J_1 \cup J_2)| \\ &= n + t - 2(a_r + 1) + k - |S_1 \cap J_1| - |S_2 \cap J_2| \\ &\geq n + t - 2a_r + k - 5 + \eta. \end{aligned} \quad (4.26)$$

Additionally, because  $(J_1 \cap S_3) \sqcup (J_2 \cap S_4) \subseteq N(a_r)$ , it follows that

$$|J_1 \cap S_3| + |J_2 \cap S_4| \leq |N(a_r)| \leq 2. \quad (4.27)$$

Lastly, because  $(J_1 \cap S_3) \cap (J_2 \cap S_4) = J_1 \cap J_2 = \emptyset$ , from Equations (4.25), (4.26), and (4.27),

$$\begin{aligned}
 |T| &\geq |T \cap (S_3 \cup S_4 \cup J_1 \cup J_2)| \\
 &= |T \cap (S_3 \cup S_4)| + |T \cap (J_1 \cup J_2)| - |T \cap (S_3 \cup S_4) \cap (J_1 \cup J_2)| \\
 &\geq |T \cap (S_3 \cup S_4)| + |T \cap (J_1 \cup J_2)| - |(S_3 \cup S_4) \cap (J_1 \cup J_2)| \\
 &\geq (n + t - \eta - 3) + (n + t - 2a_r + k - 5 + \eta) - (|J_1 \cap S_3| + |J_2 \cap S_4|) \\
 &\geq 2n + 2t + k - 8 - 2a_r - 2 = 2n + 2t + k - 2a_r - 10.
 \end{aligned} \tag{4.28}$$

Because  $|T| = \text{range}(L) + 1 - n - t$ , from Equation (4.28),  $\text{range}(L) \geq 3n + 3t + k - 2a_r - 11$ .  $\square$

Now, we strengthen Corollary 4.12.

**Lemma 4.14.** *If  $\text{isd}(P_n) = 2n - 4$  and  $n \geq 13$ , then  $T \subseteq M$ .*

**Proof.** Suppose otherwise. Because  $P_n$  is a integral sum graph of  $-L$ , by applying Corollary 4.12 on  $L$  and  $-L$ , we have  $|T \setminus M| = \xi = \eta = 1$ ,  $t = 0$ , and  $-2a_r \in L$ . Additionally, assume without loss of generality that  $a_n \geq |a_1|$ . First, from Lemma 4.2,

$$|([-2a_r, -a_r] \cup [a_r, 2a_r]) \setminus L| = k \leq 1. \tag{4.29}$$

Thus,  $|\{2a_r - 1, -2a_r + 1\} \cap S| \geq 1$ . Without loss of generality, let  $2a_r - 1 \in L$ . Therefore, because  $N(2a_r - 1) \neq \emptyset$ , and  $a_n \geq |a_1|$ , we have  $a_n \geq 3a_r - 1$ .

Now, suppose that  $k = 0$ . Because  $a_n \geq 3a_r - 1 \geq 2a_r + 2$  for  $a_r \geq 3$ , by setting  $[a, b] = [-2a_r, -a_r]$  and  $[c, d] = [a_r, 2a_r]$  in Lemma 4.7,

$$[2a_r + 2, 4a_r - 2] \subseteq T. \tag{4.30}$$

If  $a_r \geq 5$ , then from Equation (4.30),  $\{2a_r + 2, 2a_r + 3\} \subseteq T \setminus M$ , so  $a_r \leq 4$ .

If  $a_r = 4$ , from Equation (4.30),  $\{10, 14\} \subseteq T$ . Thus,  $T \setminus M = \{10\}$  because  $|T \setminus M| = 1$ . Now,  $\{9, 15\} \subseteq L$  because  $\{9, 11\} \cap T \setminus M = \emptyset$ . Then  $\{-8, -7, -6\} \subseteq N(15)$ . Thus,  $a_r \leq 3$ .

If  $a_r = 3$ , from Equation (4.30),  $\{8, 10\} \subseteq T$ . First, if  $7 \in L$ , then  $N(7) = \{-3, -4\}$  because  $\Delta = 2$ . Thus,  $\{8, 13\} \subseteq T$  because  $[4, 6] \cap N(7) = \emptyset$  and  $\{8, 10\} \subseteq T$ . Thus,  $\{8, 9\} \subseteq T \setminus M$ . So,  $7 \in T$ . Now, because  $|T \setminus M| = 1$  and  $10 \in T$ , it follows that  $T \setminus M = \{7\}$ . Thus,  $\{11, 13\} \subseteq L$  because  $\{8, 10\} \cap T \setminus M = \emptyset$ . As a result,  $\{3, 11, 12\} \subseteq N(-6)$ . Thus,  $a_r \leq 2$ . Now, because  $\xi = 1$  and  $a_r \leq 2$ , from Lemma 4.13,  $\text{range}(L) = 2n - 4 \geq 3n - 16$ , so  $n \leq 12$ .

Next, suppose that  $k = 1$ . First, because  $|T \setminus M| = 1$ , we have  $T \setminus M \subseteq [-a_r, a_r]$ . Now, if  $n \geq 4$ , then because  $a_n \geq 3a_r - 1 \geq 2a_r + 3$ , by Lemma 4.7 on  $[-2a_r, -a_r]$  and  $[a_r, 2a_r]$  with  $\tau = 1$ , we have  $[2a_r + 3, 4a_r - 3] \subseteq T$ . Now, because  $\{a_r + 3, 2a_r + 3, 3a_r - 3, 4a_r - 3\} \cap T \setminus M = \emptyset$ , it must be that  $\{a_r + 3, 3a_r + 3, 3a_r - 3, 5a_r - 3\} \subseteq L$ , and  $\{a_r, 3a_r + 3, 5a_r - 3\} \subseteq N(-2a_r)$ . Thus,  $n \leq 3$ , and it follows that  $n \leq 12$  by Lemma 4.13, because  $k = \xi = 1$ .  $\square$

We now further strengthen Corollary 4.12.

**Lemma 4.15.** *If  $\text{isd}(P_n) = 2n - 4$  for  $n \geq 13$ , then  $\eta = 1$  and  $\xi = 0$ .*

**Proof.** Suppose otherwise. Then, from Corollary 4.12 and Lemma 4.14,  $T \subseteq M$ ,  $\xi = 1$ , and  $t = \eta = 0$ . Then  $a_{r-1} < -a_r \leq -1$ . Thus,  $-1 \in T \subseteq M$ , which implies  $a_{r-1} = -(a_r + 1)$ . Therefore, as  $T \subseteq M$ , from Lemma 4.2,

$$[-2a_r, -a_r - 1] \cup [a_r, 2a_r] \subseteq L. \tag{4.31}$$

Because  $t = 0$ , it follows that  $N(2a_r - 1) \neq \emptyset$ . Therefore,  $a_1 \leq -3a_r$  or  $a_n \geq 3a_r - 1$  or both. First if  $a_n \geq 3a_r - 1$ , then from Equation (4.31) and Lemma 4.7 with intervals  $[-2a_r, -a_r]$ , and  $[a_r, 2a_r]$  and  $\tau = 0$ , it holds that  $[2a_r + 3, 4a_r - 2] \cap L = \emptyset$ .

Now, if  $a_r \geq 4$ , then  $a_n \geq 3a_r - 1 \geq 2a_r + 3$ , so  $[2a_r + 3, 4a_r - 2] \subseteq T$ . Then, because  $T \subseteq M$ , we have  $\{a_r + 3, 3a_r + 3, 3a_r - 3, 5a_r - 3, 3a_r - 2, 5a_r - 2\} \subseteq L$ . Thus,  $\{a_r + 3, 3a_r - 3, 3a_r - 2\} \subseteq N(2a_r)$ . Therefore,  $a_r \leq 3$ , so by Lemma 4.13,  $n \leq 12$  because  $k = 1$  and  $a_{r-1} = -(a_r + 1)$ . As a result,  $a_1 \leq -3a_r$ .

Now, if  $a_r \geq 4$  then  $a_1 \leq -3a_r \leq -2a_r + 3$ , so from Equation (4.31) and Lemma 4.7 with intervals  $[-2a_r, -a_r] \cup [a_r, 2a_r]$  and  $\tau = 0$ , we have  $[-4a_r + 2, -2a_r - 3] \subseteq T$ . Then, because  $T \subseteq M$ , then  $\{a_r + 3, 3a_r - 3, 3a_r - 2\} \subseteq N(2a_r)$ . Therefore,  $a_r \leq 3$ , so by Lemma 4.13,  $n \leq 12$ , because  $k = 1$  and  $a_{r-1} = -(a_r + 1)$ .  $\square$

Now, from Lemmas 4.14 and 4.15 and Corollary 4.12 if  $\text{isd}(P_n) = 2n - 4$  for  $n \geq 13$ , then  $\xi = t = 0$ ,  $\eta = 1$ , and  $T \subseteq M$ . Next, we show that if  $\eta = 1$  and  $T \subseteq M$ , then the following range of integers belong to  $L$ .

**Proposition 4.16.** *If  $\eta = 1$  and  $T \subseteq M$ , then  $[-2a_r + 1, -a_r] \cup [a_r, 2a_r - 1] \subseteq L$ .*

**Proof.** First, because  $\eta = 1$ , it holds that  $a_{r-1} = -a_r$ . Thus,  $-a_r \in L$ . In addition, by assumption,  $|T \setminus M| = 0$ . Thus, by Lemma 4.2,  $[-2a_r + 1, -a_r - 1] \cup [a_r, 2a_r - 1] \subseteq L$ .  $\square$

Now, we show through a series of lemmas that if the conditions of Proposition 4.16 are met and in addition,  $\xi = t = 0$ ,  $-2a_1 \notin L$ , and  $\text{range}(L) \geq n + 9 \geq 22$ , then  $a_r = 1$ . We begin by showing that  $a_r \leq 3$  and improve that bound to  $a_r = 1$ . While the assumptions that  $-2a_1 \notin L$  and  $\text{range}(L) \geq n + 9 \geq 22$  are not needed to show that  $a_r \leq 3$ , the assumptions that  $-2a_1 \notin L$  and  $n \geq 13$  are necessary to prove that  $a_r \leq 2$ , and the assumption that  $\text{range}(L) \geq n + 9$  is additionally needed to prove that  $a_r = 1$ .

**Lemma 4.17.** *If  $\xi = t = 0$ ,  $\eta = 1$ , and  $T \subseteq M$ , then  $a_r \leq 3$ .*

**Proof.** Suppose otherwise. First, because  $P_n$  is an integral sum graph of  $-L$ , assume that  $a_{n+t} = a_n \geq |a_1|$ . From Proposition 4.16,  $2a_r - 1 \in L$ . Then, because  $t = 0$ , it holds that  $N(2a_r - 1) \neq \emptyset$ .

Now, suppose that  $a_n \leq 3a_r - 2$ . Then  $2a_r - 1$  cannot neighbor a positive label, so it must neighbor a negative label. However, as  $3a_r - 2 \geq a_n \geq |a_1|$ , the sum of  $2a_r - 1$  and the said neighbor must be at least  $(2a_r - 1) + (-3a_r + 2) = -a_r + 1$ . However, because  $-a_r$  is the smallest negative label in  $L$ , we reach a contradiction. Thus,  $a_n \geq 3a_r - 1$ .

Next, recall that  $a_r \geq 4$  by the supposition. As a result,  $a_n \geq 3a_r - 1 \geq 2a_r + 2$ . Now, set  $(a, b, c, d) = (-2a_r + 1, -a_r, a_r, 2a_r - 1)$  in Lemma 4.7. Then by Proposition 4.16,  $\tau = 0$ . Now, from Lemma 4.7 and because  $a_r \geq 4$ , it holds that  $3a_r \in [2a_r + 2, 4a_r - 4] \subseteq T$ . Now, because  $\xi = 0$ , it holds that  $2a_r \in T$ . Thus,  $2a_r \in T \setminus M$ . Furthermore, because  $T \subseteq M$ , we reach a contradiction. Therefore,  $a_r \leq 3$ .  $\square$

**Lemma 4.18.** *If  $\xi = t = 0$ ,  $\eta = 1$ ,  $T \subseteq M$ ,  $-2a_r \notin L$  and  $n \geq 13$ , then  $a_r \leq 2$ .*

**Proof.** Suppose otherwise. By Lemma 4.17,  $a_r = 3$ . In addition, by Proposition 4.16,  $[-5, -3] \cup [3, 5] \subseteq L$ .

Next, because  $t = 0$ , it holds that  $N(5) \neq \emptyset$ . In addition, because  $P_n$  is a integral sum graph of  $-L$ , assume without loss of generality that  $a_n \geq |a_1|$ . Now, suppose that  $a_n \leq 7$ . Then 5 cannot neighbor a positive label, so it must neighbor a negative label. However, as  $7 \geq a_n \geq |a_1|$ , the sum of 5 and the said neighbor must be at least  $5 + (-7) = -2$ . However, because  $-3$  is the smallest negative label in  $L$ , we reach a contradiction. Thus,  $a_n \geq 8$ .

Now, because  $\xi = 0$ , it holds that  $6 \in T$ . In addition, by the assumption in the statement of the lemma,  $T \subseteq M$ . Thus,  $9 \in L$ . Next, set  $(a, b, c, d) = (-5, -3, 3, 5)$  in Lemma 4.7. Then by Proposition 4.16,  $\tau = 0$ . Thus,  $8 \in T \subseteq M$  and  $11 \in L$ .

Next, because  $9 \in L$ , it follows that  $\{-4, -5\} \subseteq N(9)$ . Now,  $\{-4, -5, 9\}$  forms a cycle if  $-9 \in L$ . Thus,  $-9 \notin L$ . In addition, by assumption,  $-2a_r = -6 \notin L$ . If  $a_1 \leq -6$ , then  $-6 \in T \setminus M$ , so  $a_1 = -5$ . Now, because  $\Delta = 2$ , we have  $\{-4, -5\} = N(9)$ . Thus,  $[3, 5] \cap N(9) = \emptyset$ , and so  $[12, 14] \cap L = \emptyset$ .

Now, if  $a_n \geq 15$ , then  $[12, 14] \subseteq T \subseteq M$ , so  $\{15, 16\} \subseteq L$ . Because  $13 \notin L$ , we have that  $10 \notin M$ . As if  $10 \notin L$ , then  $10 \notin T \setminus M$ , it holds that  $10 \in L$ . As a result,  $\{4, 10, 11\} \subseteq N(5)$ , but  $\Delta = 2$ . Thus,  $a_n < 15$ . Now,  $a_n = 11$ , because  $[12, 14] \cap L = \emptyset$ .

To collate,  $(a_1, a_n) = (-5, 11)$ , and  $[-2, 2] \cap L = \emptyset$ . Thus,  $n \leq 12$ . Now, from the assumption that  $n \geq 13$ , it holds that  $a_r \leq 2$ .  $\square$

**Lemma 4.19.** *If  $\xi = t = 0$ ,  $\eta = 1$ ,  $T \subseteq M$ ,  $-2a_r \notin L$ , and  $\text{range}(L) \geq n + 9 \geq 22$ , then  $a_r = 1$ .*

**Proof.** Suppose otherwise. By Lemma 4.18,  $a_r = 2$ . Thus, by Proposition 4.16,  $3 \in L$ . For  $1 \leq i \leq n - 2$ , if  $b_i \neq 1$  and  $b_{i+1} \neq 1$ , then  $\{a_{i+1} - 1, a_{i+1} + 1\} \cap T \setminus M \neq \emptyset$ . But  $T \subseteq M$  by the assumption. Thus, for  $2 \leq i \leq n - 2$ , if  $b_i = 2$ , then at least one of  $b_{i-1}$  or  $b_{i+1}$  is 1. As a result, if  $b_i = 2$  for  $2 \leq i \leq n - 2$ , then  $\{a_i, a_{i-1}\} \cap N(3) \neq \emptyset$ . Thus,  $c_2 + c_3 \leq 3$ , because  $\Delta = 2$  and it is possible that  $6 \in L$ .

Next,  $\{a_i + 1, a_i + 3\} \cap T \setminus M \neq \emptyset$  if  $b_i \geq 4$  and  $1 \leq i (\neq r - 1) \leq n - 2$ . Now, because  $T \subseteq M$  by assumption,  $b_i \leq 3$  for all  $1 \leq i (\neq r - 1) \leq n - 2$ . Next, because  $\eta = 1$  by assumption,  $b_{r-1} = 4$ . Thus,  $c_4 = 1$  and  $c_j = 0$  for all  $j \geq 5$ . Therefore,

$$\text{range}(L) = 4c_4 + 3c_3 + 2c_2 + c_1 = 4c_4 + 3c_3 + 2c_2 + (n - 1 - c_2 - c_3 - c_4) \leq n + 8. \quad (4.32)$$

Finally, by the assumption that  $\text{range}(L) \geq n + 9$ , it holds that  $a_r = 1$ .  $\square$

Now, we show that  $\text{isd}(P_n) \geq 2n - 3$  for  $n \geq 8$ .

**Lemma 4.20.** *If  $n \geq 8$ , then  $\text{isd}(P_n) \geq 2n - 3$ .*

**Proof.** Suppose otherwise. From Lemma 4.11,  $\text{isd}(P_n) = 2n - 4$ . By exhaustive computer search,  $n \geq 13$ . Because  $P_n$  is an integral sum graph labeling of  $-L$ , assume that  $a_n \geq |a_1|$ . In addition, from Lemmas 4.5, 4.14 and 4.15 and Corollary 4.12 on  $L$  and  $-L$ , we have that  $\xi = t = 0$ ,  $\eta = 1$ ,  $T \subseteq M$ ,  $|N(a_r)| = 1$ , and  $-2a_r \notin L$ . Furthermore,  $\text{range}(L) = 2n - 4 \geq n + 9$ , because  $n \geq 13$ . Thus, from Lemma 4.19,  $a_r = 1 = |a_{r-1}|$ . In addition, because  $n \geq 13$  and  $a_n \geq |a_1|$ , it follows that  $a_n \geq 11$ .



Next, if  $b_i \geq 3$ , then  $\{a_i + 1, a_i + 2\} \cap T \setminus M \neq \emptyset$ . Therefore,  $b_i \leq 2$  for  $1 \leq i \leq n - 1$ . Furthermore,  $c_1 = 2$ , because  $|N(1)| = 2$ . Thus, let  $b_j = b_k = 1$  with  $j < k$ .

Now, because  $\xi = 0$ , it holds that  $2 \in T \subseteq M$ . Then, because  $a_n \geq 11$ , it holds that  $3 \in L$ . If  $k \neq n - 1$  and  $k - j = 1$ , then  $\{-3, -1, 1\} \in N(a_j + 1)$ . However, if  $k \neq n - 1$  and  $k - j \neq 1$ , then  $\{-1, a_j + 1, -3, a_k + 1\}$  form a component. Therefore,  $k = n - 1$ .

Likewise, if  $-3 \in L$ , then  $j = 1$ . Thus, because  $a_n \geq 11$  and  $k = n - 1$ , it follows that  $\{1, 3, 5\} \subseteq L$ . Because  $j = 1$  and  $k = n - 1$ , it holds that  $\{1, 3, 5\} \subseteq N(a_1)$ . Therefore,  $-3 \notin L$ , so  $a_1 = -1$ . As  $\text{range}(L) = 2n - 4$  and  $k = n - 1$ , we have  $\{2n - 5, 2n - 6\} \subseteq L$ .

First, suppose that  $2n - 7 \in L$ . Then  $N(2n - 5) = \{-1\}$ , and  $N(2n - 7) = \{1\}$  because  $\min(L) = a_1 = -1$  and  $2 \notin L$ . Thus, because  $\Delta = 2$ , and  $\{-1, 1\} \subseteq N(2n - 6)$ , it follows that  $\{-1, 1, 2n - 7, 2n - 6, 2n - 5\}$  forms a component. Therefore,  $2n - 7 \notin L$ . As a result,  $2n - 8 \in L$ , otherwise  $2n - 8 \in T \setminus M$ . Because  $\min(L) = -1$  and  $2n - 7 \notin L$ , we have  $|N(2n - 5)| = |N(2n - 6)| = 1$ . Now,  $|N(2n - 8)| = 2$ , because  $G = P_n$ . Therefore,  $j = n - 3$ . Then,  $\{-1, 1, 3, 2n - 9, 2n - 8, 2n - 6, 2n - 5\}$  forms a component. Thus,  $\text{isd}(P_n) \geq 2n - 3$ .  $\square$

Now, a better lower bound for  $\text{ispum}(P_n)$  follows from Lemma 4.20.

**Theorem 4.21.** For  $n \geq 8$ , it holds that

$$2n - 3 \leq \text{ispum}(P_n) \leq \begin{cases} 2n - 3 & \text{if } n \text{ is even} \\ \frac{5}{2}(n - 3) & \text{if } n \text{ is odd.} \end{cases}$$

**Proof.** From [4, Proposition 7.2],  $\text{isd}(P_n) \leq \text{ispum}(P_n)$ . Thus, the lower and upper bounds of the statement of the theorem follow from Lemma 4.20 and [5, Theorem 7.3], respectively.  $\square$

#### 4.3. The inequality $\text{isd}(P_n) \geq 2n - 2$ for odd $n \geq 27$

Now, we show that for odd  $n$ ,  $\text{isd}(P_n) \neq 2n - 3$ . First, we state the following corollary of Lemma 4.5.

**Corollary 4.22.** If  $\text{isd}(P_n) = 2n - 3$ , then  $2t + |T \setminus M| + 2 = \xi + \eta + |N(a_r)|$ .

Next, we show that if  $\text{isd}(P_n) = 2n - 3$ , then  $L$  has no isolated vertices.

**Lemma 4.23.** If  $\text{isd}(P_n) = 2n - 3$  for  $n \geq 10$ , then  $t = 0$ .

**Proof.** Suppose otherwise. Then, as  $P_n$  is a integral sum graph of  $-L$ , by applying Corollary 4.22 to  $L$  and  $-L$ , we have  $T \subseteq M$ ,  $|N(a_r)| = 2$ ,  $\xi = \eta = t = 1$ , and  $-2a_r \in L$ . Thus, from Lemmas 4.8 and 4.10,  $a_r = 1$ , and  $2n - 3 = \text{isd}(P_n) \geq 3n - 12$ , so  $n \leq 9$ .  $\square$

We now further strengthen Corollary 4.22 by showing that  $\xi = 0$ .

**Lemma 4.24.** If  $\text{isd}(P_n) = 2n - 3$  for  $n \geq 27$ , then  $\xi = 0$ .

**Proof.** Suppose otherwise. First, if  $a_r \leq 9$ , from Lemma 4.13,  $n \leq 26$ . Thus,  $a_r \geq 10$ , and from Corollary 4.22,  $|T \setminus M| \leq 2$ . Then, from Lemma 4.2,

$$|[a_{r-1} - a_r + 1, -a_r - 1] \cup [a_r, 2a_r] \setminus L| \leq 2. \quad (4.33)$$

From Equation (4.33), there exists  $\ell \in \{2a_r - 1, -2a_r + 1, 2a_r - 2\}$  in  $L$ . Then, because  $t = 0$ , and  $N(\ell) \neq \emptyset$ , we have  $\max(|a_1|, a_n) \geq 3a_r - 2$ .

First suppose that equality holds in Equation (4.33). From Lemma 4.7, with  $\tau = 2$ , and intervals  $[a_{r-1} - a_r + 1, -a_r - 1] \cup [a_r, 2a_r]$ , and because  $\max(|a_1|, a_n) \geq 3a_r - 2 \geq 2a_r + 5$ , we have  $[-3a_r + a_{r-1} + 5, -2a_r - 5] \subseteq T$  or  $[2a_r + 5, 3a_r - a_{r-1} - 5] \subseteq T$  or both. Then  $2a_r + 5 \in T \setminus M$  or  $-2a_r - 5 \in T \setminus M$  or both. But,  $|T \setminus M \cap [a_{r-1}, a_r]| = 2$  and  $|T \setminus M| \leq 2$ . Thus,  $|[a_{r-1} - a_r + 1, -a_r - 1] \cup [a_r, 2a_r] \setminus L| \leq 1$ .

Next, suppose  $|[a_{r-1} - a_r + 1, -a_r - 1] \cup [a_r, 2a_r] \setminus L| = 1$ . From Lemma 4.7 with  $\tau = 1$ , and intervals  $[a_{r-1} - a_r + 1, -a_r - 1] \cup [a_r, 2a_r]$ , and because  $\max(|a_1|, a_n) \geq 3a_r - 2 \geq 2a_r + 4$ , it must be that  $[-3a_r + a_{r-1} + 4, -2a_r - 4] \subseteq T$  or  $[2a_r + 4, 3a_r - a_{r-1} - 4] \subseteq T$ . As a result,  $\{|-2a_r - 4, -2a_r - 5, 2a_r + 4, 2a_r + 5\} \cap T \setminus M| \geq 2$ . However,  $|T \setminus M \cap [a_{r-1}, a_r]| = 1$  and  $|T \setminus M| \leq 2$ . Thus,  $|[a_{r-1} - a_r + 1, -a_r - 1] \cup [a_r, 2a_r] \setminus L| = 0$ .

Then, from Lemma 4.7, with  $\tau = 0$  and intervals  $[a_{r-1} - a_r + 1, -a_r - 1] \cup [a_r, 2a_r]$ , we have  $([-3a_r + a_{r-1} + 3, -2a_r - 3] \cup [2a_r + 3, 3a_r - a_{r-1} - 3]) \cap L = \emptyset$ . As a result, and also because  $\max(|a_1|, a_n) \geq 3a_r - 2 \geq 2a_r + 4$ , it must be that  $[2a_r + 3, 2a_r + 5] \subseteq T \setminus M$  or  $[-2a_r - 3, 2a_r - 5] \subseteq T \setminus M$  or both. Therefore,  $\xi = 0$ .  $\square$

We now show if  $\text{isd}(P_n) = 2n - 3$  for  $n \geq 27$ , then  $|N(a_r)| = 2$ .

**Lemma 4.25.** *If  $\text{isd}(P_n) = 2n - 3$  for  $n \geq 27$ , then  $|N(a_r)| = 2$ .*

**Proof.** Suppose otherwise. By Lemma 4.23,  $t = 0$ , and so  $|N(a_r)| = 1$ . Now, Lemma 4.24,  $\xi = 0$ . Next, by Corollary 4.22,  $|T \setminus M| = \eta - 1 \geq 0$ . Thus,  $T \subseteq M$  and  $\eta = 1$ . Now, observe that  $P_n$  is an integral sum graph of  $-L$ . Because  $\xi = 0$  and  $\eta = 1$ , by Lemma 4.24,  $-2a_1 \notin L$ . In addition, because  $n \geq 27$  and  $\text{isd}(P_n) = 2n - 3$ , it holds that  $\text{range}(L) \geq n + 9 \geq 22$ . Thus, by Lemma 4.19,  $a_r = 1$ .

Now, by applying Corollary 4.22 on  $-L$ , it holds that  $|N(a_{r-1})| = 1$ . Thus, because  $G = P_n$  and  $t = 0$ , if  $a_i \notin \{1, -1\}$  then  $|N(a_i)| = 2$ . In addition, because  $|N(a_r)| = 1$  and  $T \subseteq M$ , we have  $c_1 = 1$ , and  $c_2 = n - 2$ . Thus, one of  $a_1, a_n$  is even. Without loss of generality, let  $a_n$  be even. Since  $1 \in L$ , all even labels are in  $S_2$ , and all labels in  $S_1$  are odd.

However, as  $a_n = \max(L)$ , it follows that  $N(a_n) \subseteq S_1$ . Let  $N(a_n) = \{a_j, a_k\}$  with  $j < k$ . Then, because  $\{a_j, a_k + a_n\} \subseteq L$ , all odd labels between  $a_j$  and  $a_k + a_n$  must be in  $L$ . Thus, because  $a_j \leq a_k$ , it follows that  $\{a_j, a_j + 2, a_k + 2\} \subseteq N(a_n - 2)$ . As a result,  $a_n - 2 \notin L$  and  $a_n$  is the only even label. Then,  $|N(a_i)| \leq 1$  for  $1 \leq i \leq n - 2$ , so  $|N(a_r)| = 2$ .  $\square$

We now show that if  $\text{isd}(P_n) = 2n - 3$  for  $n \geq 27$ , then  $|T \setminus M| = \eta = 1$ .

**Lemma 4.26.** *If  $\text{isd}(P_n) = 2n - 3$  for  $n \geq 27$ , then  $|T \setminus M| = \eta = 1$ .*

**Proof.** Suppose otherwise. From Corollary 4.22 and Lemmas 4.23, 4.24, and 4.25,  $|T \setminus M| = \eta = 0$ . Thus,  $a_{r-1} = -(a_r + 1)$  because  $|a_{r-1}| \neq a_r$  and  $-1 \in T \subseteq M$ . Now, from Lemma 4.2,

$$([-2a_r, -1 - a_r] \cup [a_r, 2a_r - 1]) \subseteq L. \quad (4.34)$$

Next, from Lemma 4.7 with  $\tau = 0$  and intervals  $[-2a_r, -1 - a_r] \cup [a_r, 2a_r - 1]$ ,

$$([-4a_r + 3, -2a_r - 3] \cup [2a_r + 3, 4a_r - 3]) \cap L = \emptyset. \quad (4.35)$$

First, because  $N(-2a_r) \neq \emptyset$  from Lemma 4.23 and Equation (4.34),  $\max(|a_1|, a_n) \geq 3a_r$ . Now, if  $a_r \geq 6$ , because  $\max(|a_1|, a_n) \geq 3a_r \geq 2a_r + 3$ , it must be that  $[-4a_r + 3, -2a_r - 3] \subseteq T$  or  $[2a_r + 3, 4a_r - 3] \subseteq T$  or both. Thus,  $|[-2a_r - 3, 2a_r + 3] \cap T \setminus M| \geq 1$ . Therefore,  $a_r \leq 5$ . Finally, because  $k \geq 1$  from  $-a_r \notin L$ , and  $-2a_r \in L$  from Equation (4.34), it follows from Lemma 4.13 that  $2n - 3 = \text{isd}(P_n) \geq 3n - 2a_r - 10$ , so  $n \leq 17$ .  $\square$

Now, we use a series of lemmas to prove that if  $\text{isd}(P_n) = 2n - 3$  for  $n \geq 27$ , then  $a_r = 1$ . We start by showing that  $a_r \leq 4$  and subsequently improve the bound to  $a_r = 1$ .

**Lemma 4.27.** *If  $\text{isd}(P_n) = 2n - 3$  for  $n \geq 27$ , then  $a_r \leq 4$ .*

**Proof.** Suppose otherwise. From Lemma 4.26,  $|T \setminus M| = \eta = 1$ . Thus, from Lemma 4.2,

$$|([-2a_r + 1, -a_r] \cup [a_r, 2a_r - 1]) \setminus L| \leq 1. \quad (4.36)$$

As  $t = 0$  by Lemma 4.23, and  $2a_r - 1 \in S$  or  $-2a_r + 1 \in S$  or both, and because  $P_n$  is a sum graph labeling of  $-L$ , assume without loss of generality that  $\max(|a_1|, a_n) = a_n \geq 3a_r - 1$ . In addition, because  $3a_r - 1 > 2a_r + 3$ , from Equation (4.36) and by setting  $\tau = 1$ , with intervals  $[-2a_r + 1, -a_r] \cup [a_r, 2a_r - 1]$  in Lemma 4.7,  $[2a_r + 3, 4a_r - 5] \subseteq T$ . Thus,  $3a_r \in T$ . However, from Lemmas 4.24 and 4.26,  $2a_r = T \setminus M$ . Therefore, because  $[a_{r-1}, a_r] \cap T \setminus M = \emptyset$ ,

$$[-2a_r + 1, -a_r] \cup [a_r, 2a_r - 1] \subseteq L, \quad (4.37)$$

so because  $a_n \geq 3a_r - 1 \geq 2a_r + 2$ , by setting  $\tau = 0$  and disjoint intervals  $[-2a_r + 1, -a_r] \cup [a_r, 2a_r - 1]$  in Lemma 4.7,  $[2a_r + 2, 4a_r - 4] \subseteq T$ . For  $a_r \geq 6$ , it holds that  $\{2a_r, 2a_r + 2\} \subseteq T \setminus M$ . Thus,  $a_r = 5$ , so  $\{2a_r + 2, 4a_r - 4\} = \{12, 16\} \in T$ . Then because  $\{11, 12\} \cap T \setminus M = \emptyset$ , we have  $11, 17 \in L$ , and thus,  $\{-5, -6, 6\} \subseteq N(11)$ . Therefore,  $a_r \leq 4$ .  $\square$

**Lemma 4.28.** *If  $\text{isd}(P_n) = 2n - 3$  for  $n \geq 27$ , then  $a_r \leq 3$ .*

**Proof.** Suppose otherwise. Then by Lemmas 4.26 and 4.27,  $a_r = 4$ , and  $|T \setminus M| = \eta = 1$ . Thus, from Lemma 4.2,  $|([-7, -4] \cup [4, 7]) \setminus L| \leq 1$ . From Lemma 4.23 and because  $|\{7, -7\} \cap L| \geq 1$ , it follows that  $N(7) \neq \emptyset$  or  $N(-7) \neq \emptyset$  or both. Therefore, because  $P_n$  is a sum graph labeling of  $-L$ , we have  $\max(|a_1|, a_n) = a_n \geq 11$ .

First, suppose that  $[-7, -4] \cup [4, 7] \subseteq L$ . Because  $a_n \geq 11$ , by setting  $\tau = 0$  and disjoint intervals  $[-7, -4]$  and  $[4, 7]$  in Lemma 4.7,  $[10, 12] \subseteq T$ . Then, because  $8 \in T$  from Lemma 4.24, and  $|T \setminus M| = 1$ , it holds that  $T \setminus M = \{8\}$ . Thus, as  $\{10, 11, 12\} \cap T \setminus M = \emptyset$ , we have  $\{10, 11, 12\} \subseteq M$ , so  $[14, 16] \subseteq L$ . Now,  $9 \notin L$ , otherwise  $\{5, 6, 7\} \subseteq N(9)$ . Thus, because

$9 \notin T \setminus M$ , we have  $13 \in L$ . Because  $|N(13)| \leq 2$  and  $\{-7, -6\} \subseteq N(13)$ , we must have  $-8 \notin N(13)$ , and thus  $-8 \notin L$ . However, if  $-8 \in T$ , as  $-8 \notin T \setminus M$ , we have that  $-12 \in L$ , which implies that  $\{5, 6, 7\} \subseteq N(-12)$ . Therefore,  $-8 \notin T$  so  $a_1 = -7$ , and as  $n \geq 27$ , we have that  $a_n \geq 20$ . As  $[4, 7] \cap N(13) = \emptyset$  and  $a_n \geq 20$ , it follows that  $[17, 20] \subseteq T$ . Because  $17 \notin T \setminus M$ , it holds that  $21 \in L$ . However, then  $\{-7, -6, -5\} \subseteq N(21)$ .

Thus,  $[-7, -4] \cup [4, 7] \not\subseteq L$ . Now, from Lemma 4.7 with  $\tau = 1$  and disjoint intervals  $[-7, -4]$  and  $[4, 7]$ , and because  $a_n \geq 11$ , it holds that  $11 \in T$ , so  $15 \in L$ . Furthermore, from Lemma 4.24,  $8 \in T$ , and thus,  $12 \in L$ . We now consider each of  $([-7, 4] \cup [4, 7]) \setminus L$ .

First, because  $\eta = 1$ , we have  $\{-4, 4\} \subseteq L$ . Now,  $6 \notin T$ , as otherwise  $10 \in L$ , and  $\{-5, -6, 5\} \subseteq N(10)$ . Similarly,  $-6 \notin T$ , as otherwise  $-10 \in L$ , so  $\{5, 6, 15\} \subseteq N(-10)$ . Next, suppose that  $5 \in T$ . Then  $9 \in L$ , so  $\{9, 12\} \subseteq N(-5)$ . As  $N(-5) \leq 2$ , we have  $\{14, 15\} \cap N(5) = \emptyset$ , so  $\{10, 14\} \subseteq T$ , and thus,  $10 \in T \setminus M$ . Thus,  $5 \notin T$ . Now, suppose that  $-5 \in T$ . Then,  $-9 \in L$ , which implies  $-8 \in T$ , because  $-8 \notin L$  by applying Lemma 4.24 on  $-L$ . Thus,  $\{-12, -9, 7\} \subseteq N(5)$ . Therefore,  $-5 \notin T$ .

Additionally, if  $7 \in T$ , because  $11 \in T$ , we have  $7 \in T \setminus M$ , a contradiction. Note that  $-7 \notin T$  as otherwise  $-11 \in L$  and  $\{5, 6, 7\} \subseteq N(-11)$ . Therefore, if  $-7 \notin L$ , then  $a_1 = -6$ . However, as  $\{-4, -5, 6\} \subseteq N(9)$  and  $\{-4, -5, -6\} \subseteq N(10)$ , it follows that  $\{9, 10\} \subseteq T$ , so  $\{13, 14\} \subseteq L$ . Then, because  $|N(7)| \leq 2$ , and  $\{5, 6\} \subseteq N(7)$ , it follows that  $[12, 15] \cap N(7) = \emptyset$ , so  $[19, 22] \cap L = \emptyset$ . Now, as  $15 \in L$  and  $[9, 11] \subseteq T$ , we have  $\{-4, -5, -6\} \cap N(15) = \emptyset$ . As  $N(15) \neq \emptyset$ , it holds that  $\min(N(15)) \geq 4$ , so  $a_n \geq 19$ . However, because  $[19, 22] \cap L = \emptyset$ , it holds that  $[19, 22] \subseteq T$ , and thus,  $[16, 18] \subseteq L$ , which implies  $\{7, 12, 13\} \subseteq N(5)$ . Thus,  $a_r \leq 3$ .  $\square$

**Lemma 4.29.** If  $\text{isd}(P_n) = 2n - 3$  for  $n \geq 27$ , then  $a_r \leq 2$ .

**Proof.** Suppose otherwise. Then,  $a_r = 3$  by Lemma 4.28. Now, from Lemma 4.26,  $|T \setminus M| = \eta = 1$ . Thus, from Lemma 4.2,  $|([-5, -3] \cup [3, 5]) \setminus L| \leq 1$ .

Then, because  $P_n$  is a sum graph labeling of  $-L$ , assume without loss of generality that  $[3, 5] \subseteq L$ . First, if  $b_i = 5$ , then  $|[a_i + 1, a_i + 4] \cap T \setminus M| \geq 1$ . Because  $|T \setminus M| = 1$ , it holds that  $c_5 \leq 1$ . Similarly, if  $b_i \geq 6$  but  $i \neq r - 1$ , then  $a_i + 1, a_i + 4 \in T$  and  $a_i + 2, a_i + 5 \in T$ . Thus, because  $|T \setminus M| = 1$  and  $b_{r-1} = 6$ , we have  $c_6 = 1$ . Next, because  $[3, 5] \subseteq L$ , it follows that 4, 8 are not consecutive. In addition, from Lemma 4.24,  $6 \notin L$ . Thus, because  $|N(4)| \leq 2$  and  $|N(3)| \leq 2$ , we have  $c_4 \leq 2$ , and  $c_3 \leq 2$ .

Finally, we bound  $c_2$ . First, if  $b_1 = 2$  and  $b_{n-1} = 2$ , then  $\{a_1 + 1, a_n - 1\} \subseteq T \setminus M$ , a contradiction. Thus, assume that  $b_{n-1} \neq 2$ . Now, if  $b_i = 2$  and  $b_{i+1} = 1$ , then  $a_i \in N(3)$ . Next, if  $b_i = 2$  and  $b_{i+1} = 2$ , then  $a_i \in N(4)$ . In addition, if  $b_i = 2$  and  $b_{i+1} \geq 3$ , then  $\{a_i + 1, a_i + 4\} \cap T \setminus M \neq \emptyset$ . Now, because  $|N(3)|, |N(4)| \leq 2$ , and  $|T \setminus M| = 1$ ,

$$c_2 \leq (2 - c_3) + (2 - c_4) + (1 - c_5) = 5 - c_3 - c_4 - c_5. \quad (4.38)$$

Thus,  $\max(c_3, c_4) \leq 2$  and  $c_5 \leq 1$ , so from Equation (4.38), it holds that

$$\begin{aligned} \text{range}(L) &= \sum_{\ell=1}^{\max(b_i)} \ell \cdot c_\ell = 6 + 5c_5 + 4c_4 + 3c_3 + 2c_2 + (n - 1 - \sum_{\ell=2}^{\max(b_i)} c_\ell) \\ &= n + 4 + 4c_5 + 3c_4 + 2c_3 + c_2 \\ &\leq n + 4 + 4c_5 + 3c_4 + 2c_3 + (5 - c_3 - c_4 - c_5) \leq n + 18. \end{aligned} \quad (4.39)$$

Therefore, because  $2n - 3 = \text{range}(L) \leq n + 18$ , it holds that  $n \leq 21$ .  $\square$

**Lemma 4.30.** If  $\text{isd}(P_n) = 2n - 3$  for  $n \geq 27$ , then  $a_r = 1$ .

**Proof.** Suppose otherwise. Then, by Lemma 4.29,  $a_r = 2$ . First, if  $b_i = 4$ , with  $i \neq r - 1$ , then  $\{a_i + 1, a_i + 3\} \subseteq T$ , which results in an element of  $T \setminus M$ . Thus, because  $|T \setminus M| \leq 1$ , and  $b_{r-1} = 4$ , we have  $c_4 \leq 2$ . In addition,  $|[-3, 3] \cap L| \geq 1$ , as otherwise  $\{-1, 1\} \subseteq T \setminus M$ , a contradiction to  $|T \setminus M| = 1$ . Thus, because  $N(\pm 3) \leq 2$ , we have  $c_3 \leq 3$ . Now, if  $b_1 = b_{n-1} = 2$ , then  $\{a_1 + 1, a_n - 1\} \subseteq T \setminus M$ , a contradiction. Thus, assume that  $b_{n-1} \neq 2$ . Then, if  $b_i \neq 1$  and  $b_{i+1} \neq 1$ , then  $\{a_{i+1} - 1, a_{i+1} + 1\} \subseteq T$ , which results in an element of  $T \setminus M$ . Therefore, for each  $b_i = 2$ , either  $b_{i+1} = 1$  or  $a_i + 1 \in T \setminus M$ .

Now, because  $|T \setminus M| = 1$ , and  $|N(3)| \leq 2$ , it holds that

$$c_2 \leq (3 - c_3) + (1 - c_4) = 4 - c_3 - c_4. \quad (4.40)$$

Therefore, because  $c_3 \leq 3$  and  $c_4 \leq 2$ , it holds that

$$\begin{aligned} \text{range}(L) &\leq 4c_4 + 3c_3 + 2c_2 + (n - 1 - c_4 - c_3 - c_2) \\ &\leq n - 1 + 3c_4 + 2c_3 + (4 - c_3 - c_4) \\ &= n + 3 + 2c_4 + c_3 \leq n + 10. \end{aligned} \quad (4.41)$$

Thus,  $\text{isd}(P_n) = 2n - 3 \leq n + 10$ , from which  $n \leq 13$ , a contradiction.  $\square$

We now arrive at a lower bound on  $\text{isd}(P_n)$  for odd  $n \geq 27$ .

**Lemma 4.31.** *For odd  $n \geq 27$ , it holds that  $\text{isd}(P_n) \geq 2n - 2$ .*

**Proof.** Suppose otherwise. Then, by Lemmas 4.26 and 4.30,  $|a_{r-1}| = a_r = 1$ . In addition, from Lemmas 4.24 and 4.25, because  $|N(1)| = 2$  and  $2 \notin L$ , we have  $c_1 = 2$ . Furthermore, from Lemma 4.26,  $|T \setminus M| = 1$ , so  $c_3 = 1$ , and there is no  $i$  such that  $b_i \geq 4$ . Therefore, let  $b_f = 1$ ,  $b_g = 1$ , and  $b_h = 3$ , where  $f < g$ . In addition, for every  $i \notin \{f, g, h\}$ , we have  $b_i = 2$ .

In addition, for  $n \geq 14$ , there are at least 3 indices  $i$  for which  $b_i = b_{i+1} = b_{i+2} = 2$ . Thus,  $\{4, -4, 6, -6\} \cap L = \emptyset$ , as otherwise  $|N(\pm 6)|, |N(\pm 4)| \geq 2$ . It follows that either  $-3 \in L$  or  $\min(L) = -1$ , and either  $3 \in L$  or  $\max(L) = 1$ . In addition, because  $6 \notin L$  and  $b_h = 3$ , there is at most one index  $i$  such that  $\{b_i, b_{i+1}\} = \{1, 2\}$ , otherwise  $|N(3)| > 2$  or  $|N(-3)| > 2$ .

First, we consider  $h \notin [f, g]$ . Since  $-L$  is also a sum graph labeling of  $P_n$ , without loss of generality let  $h > g$ . If  $g \neq f + 1$ , then  $\{b_f, b_{f+1}\} = \{b_{g-1}, b_g\} = \{1, 2\}$ . Therefore,  $g = f + 1$ . Now suppose that  $f \neq 1$ . Then, because  $h > g$ , it holds that  $b_{f-1} = 2$ , which implies  $-3 \notin L$ , as otherwise  $\{-1, 1, -3\} \subseteq N(a_g)$ . Thus, it follows that  $\min(L) = a_1 = -1$ . Then because  $t = 0$  from Lemma 4.23, it holds that  $b_{n-1} = 1$ , because otherwise  $a_n$  would be isolated, for  $\min(L) = -1$ . However,  $h > g$ , which implies  $b_{n-1} \neq 1$ . Thus,  $f = 1$ . Now, if  $3 \in L$ , then  $h = g + 1$ , as otherwise  $\{-1, 1, 3\} \subseteq N(a_g)$ . However, this forces  $|N(3)| = |N(-3)| = |N(a_n)| = 1$ , a contradiction to  $G = P_n$ . Thus,  $3 \notin L$  and thus,  $1 = \max(L) = a_n$ , i.e., 1 is the only positive label of  $L$ . It follows from  $f = g - 1 = 1$  and  $2 \notin L$  that  $a_3 = a_1 + 2$  and  $a_1$  have exactly one neighbor. Therefore, because  $|N(-1)| = |N(1)| = |N(a_2)| = 2$ , it follows that  $\{-1, 1, a_1, a_2, a_3\}$  is a component.

Therefore,  $h \in [f, g]$ . First, if  $f \neq 1$  and  $g \neq n - 1$ , we have  $\{b_{f-1}, b_f\} = \{1, 2\} = \{b_g, b_{g+1}\}$ , a contradiction, because there is at most one  $i$  such that  $\{b_i, b_{i+1}\} = \{1, 2\}$ . Therefore, either  $f = 1$  or  $g = n - 1$ . Assume without loss of generality that  $g = n - 1$ . First, suppose that  $f = 1$ . Then either  $h = g - 1$  or  $h = f + 1$ , as otherwise,  $\{b_{f+1}, b_f\} = \{1, 2\} = \{b_{g-1}, b_g\}$ . Assume without loss of generality that  $h = g - 1$ . Now, because  $\{-6, -4, 6\} \cap L = \emptyset$ , we have  $|a_1|, a_n \geq 8$ . Thus,  $\{1, 3, 5, a_1 + 1, a_1 + 3, a_1 + 5\} \subseteq L$ , so  $\{1, 3, 5\} \subseteq N(a_1)$ . Therefore,  $f \neq 1$ , which forces  $f + 1 = h = g - 1$ , as otherwise  $\{b_{f-1}, b_f\} = \{1, 2\}$  and either  $\{b_f, b_{f+1}\} = \{1, 2\}$  or  $\{b_{g-1}, b_g\} = \{1, 2\}$ . Because  $b_{n-4} = 2$ , we have  $-3 \notin L$ , as otherwise  $\{-3, -1, 3\} \subseteq N(a_{n-2})$ . Therefore,  $a_1 = -1$ , and  $\{f, g, h\} = \{n - 3, n - 2, n - 1\}$ , which corresponds to the set of all odds from  $[-1, 2n - 9]$ , combined with  $\{2n - 8, 2n - 5, 2n - 4\}$ . For odd  $n$ ,  $\{n - 4, n - 2, 2n - 4\}$  each have exactly one neighbor, which contradicts  $G = P_n$ .  $\square$

We now evaluate  $\text{isd}(P_n)$  for  $n \geq 3$ .

**Theorem 4.32.** *For  $n \geq 3$ , it holds that*

$$\text{isd}(P_n) = \begin{cases} 2n - 3 & \text{if } n = 3 \\ 2n - 4 & \text{if } 4 \leq n \leq 7 \\ 2n - 3 & \text{if } 8 \leq n \leq 9 \\ 2n - 3 & \text{if } n \geq 10 \text{ is even} \\ 2n - 2 & \text{if } n \geq 11 \text{ is odd.} \end{cases}$$

**Proof.** The statement is true for  $n \leq 26$  by exhaustive computer search. Now, the lower bounds follow from Lemmas 4.20 and 4.31. The upper bounds follow from Theorem 4.1.  $\square$

## 5. The sum-diameter of paths

We prove Theorem 5.6. The current best bound on  $\text{sd}(P_n)$  is by Li.

**Theorem 5.1** ([4, Proposition 9.4]). *For  $n \geq 3$ , it holds that*

$$2n - 3 \leq \text{sd}(P_n) \leq 2n - 2.$$

For the rest of this section, let  $L$  be a sum graph labeling of a graph  $G \cup I_m$  that satisfies  $\text{range}(L) = \text{sd}(G)$ . Let  $S = \{a_1, a_2, \dots, a_n\} \subseteq \mathbb{N}$  be the labels of the vertices of  $G$  such that  $a_1 < a_2 < \dots < a_n$ . We first cite Li [4] on the labels that must be in  $S$  if  $\text{spum}(P_n) = 2n - 3$ .

**Lemma 5.2** ([4, Remark 7.8]). *If  $\text{sd}(P_n) = 2n - 3$ , then  $[a_1, 2a_1] \subseteq S$ .*

In addition, following the proof of Lemma 3.2 directly after defining  $T, M$ , and  $N$  analogously as Section 3 gives us the following lower bound on  $\text{sd}(G)$ .

**Corollary 5.3.** *If  $[a_1, 2a_1] \subseteq S$ , then  $\text{sd}(G) \geq 3n - a_1 - 4 - 4\Delta + \delta$ .*

Next, we give a lower bound on  $n$  with respect to  $a_1$  when  $\text{spum}(P_n) = 2n - 3$  and  $n - a_1 \geq 4$ .

**Lemma 5.4.** *If  $\text{sd}(P_n) = 2n - 3$  and  $n - a_1 \geq 4$ , then  $n \geq 2a_1$ .*

**Proof.** By Lemma 5.2, it holds that  $[a_1, 2a_1] \subseteq S$ . Now, because  $\text{range}(L) = 2n - 3$ , it holds that  $\max(S) \leq 2n - 3$ , otherwise  $\max(S)$  would be isolated. Therefore,

$$|S \cap [2a_1 + 1, 2n - 3]| = n - (a_1 + 1) = n - a_1 - 1. \quad (5.1)$$

As a result,

$$\begin{aligned} |S \cap [2a_1 + 1, n + a_1 + 1]| &= |S \cap [2a_1 + 1, 2n - 3 - (n - a_1 - 4)]| \\ &\geq |S \cap [2a_1 + 1, 2n - 3]| - (n - a_1 - 4) = 3. \end{aligned} \quad (5.2)$$

Thus, if  $a_1 + 2a_1 \geq n + a_1 + 1$ , then  $N(a_1) \geq 3$ . As a result,  $n \geq 2a_1$ .  $\square$

Next, we give an upper and lower bound on  $a_1$  when  $\text{range}(L) = 2n - 3$ .

**Lemma 5.5.** *If  $\text{sd}(P_n) = 2n - 3$ , then  $n - 2 \geq a_1 \geq n - 8$ .*

**Proof.** By Corollary 5.3,  $\text{sd}(P_n) = 2n - 3 \geq 3n - a_1 - 11$ . Therefore,  $a_1 \geq n - 8$ . In addition, if  $a_1 \geq n - 1$ , then  $a_n \geq a_1 + (n - 1) = 2n - 2$ . Thus,  $\text{range}(L) \geq 2n - 2 + a_1 - a_1 \geq 2n - 2$ . As a result,  $n - 2 \geq a_1$ .  $\square$

We now evaluate  $\text{sd}(P_n)$  for  $n \geq 3$ .

**Theorem 5.6.** *For  $n \geq 3$ , it holds that*

$$\text{sd}(P_n) = \begin{cases} 2n - 3 & \text{if } 3 \leq n \leq 6 \\ 2n - 2 & \text{if } n \geq 7 \end{cases}.$$

**Proof.** By exhaustive computer search, the statement of the theorem holds when  $n \leq 16$ . Now, suppose that  $n \geq 17$ . From Theorem 5.1, it suffices to show that  $\text{sd}(P_n) \neq 2n - 3$ . Thus, suppose for the sake of contradiction that  $\text{sd}(P_n) = 2n - 3$ . Then by Lemma 5.5, it holds that  $n - 2 \geq a_1 \geq n - 8$ . Now, if  $n - 4 \geq a_1 \geq n - 8$ , then it follows from Lemma 5.4 that  $n \geq 2a_1 \geq 2n - 16$ . Thus,  $n \leq 16$ , and we reach a contradiction as sought.

Now, if  $a_1 = n - 3$ , then  $\max(L) = 3n - 6$ . Because all  $s \in S$  must satisfy  $n - 3 + s \leq 3n - 6$ , it must hold that  $s \leq 2n - 3$ . In addition, because  $|S| = n$ , it must hold that

$$|[n - 3, 2n - 3] \setminus S| = 1. \quad (5.3)$$

Now, if  $2n - 3 \in S$ , then  $\{2n - 3, n\} \subseteq N(n - 3)$ . Therefore,  $\{n - 1, n - 2\} \cap N(n - 3) = \emptyset$ , and thus,  $2n - 4, 2n - 5 \notin S$ . But if so, then  $|[n - 3, 2n - 3] \setminus S| \geq 2$ . Thus  $2n - 3 \notin S$ . However, from Equation (5.3), it holds that  $S = [n - 3, 2n - 4]$ . Then,  $N(n - 2) = \{n - 3, 2n - 4\}$ , and thus  $[n - 1, 2n - 3] \notin N(n - 2)$ . Therefore  $[2n - 3, 3n - 7] \notin L$ . But then, since  $|N(2n - 5)| = |N(2n - 4)| = 1$ , we have  $\{2n - 4, n - 2, n - 3, n - 1, 2n - 5\}$  is a connected component, a contradiction as sought.

Finally, if  $a_1 = n - 2$ , then because  $\max(L) = 3n - 5$ , any  $s \in S$  should satisfy  $s \leq 3n - 5 - (n - 2) = 2n - 3$ . Because  $|S| = n$ , it must be that  $S = [n - 2, 2n - 3]$ . Then  $N(n - 2) = \{n - 1, 2n - 3\}$ , and thus  $[n, 2n - 4] \notin N(n - 2)$ , which implies  $[2n - 2, 3n - 6] \notin L$ . However, then  $\{2n - 3, n - 2, n - 1, 2n - 4\}$  is a connected component, a contradiction as sought.  $\square$

## Declaration of competing interest

None.

## Data availability

No data was used for the research described in the article.

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